

Regulation and Competition: Implications for Efficiency and Stability in the Banking Industry *

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August 3, 2026

Abstract

There is substantial size heterogeneity in the US banking industry. The top 4 banks out of 4000 comprise over 40% of the market share. Other countries are even more concentrated. Here, I consider the implications of heterogeneity in the banking industry for efficiency and stability of that sector along with the conduct of regulatory and monetary policy. Quantitative dynamic models of the banking industry can be used to evaluate regulation to consider possible tradeoffs between allocative efficiency and financial stability that are absent in widely used representative bank models. The framework integrates micro- and macro-prudential policy evaluation; unlike small (measure zero) banks, an idiosyncratic shock to a big bank with market power can spill over to the aggregate economy as in the granularity literature. This chapter attempts to bring the micro and macro literatures on banking together in a unified framework.

Keywords: Systemic Risk, Macro-Prudential Policy, Granularity, Banking Industry Dynamics

JEL Classification Numbers: G21, E58, D43.

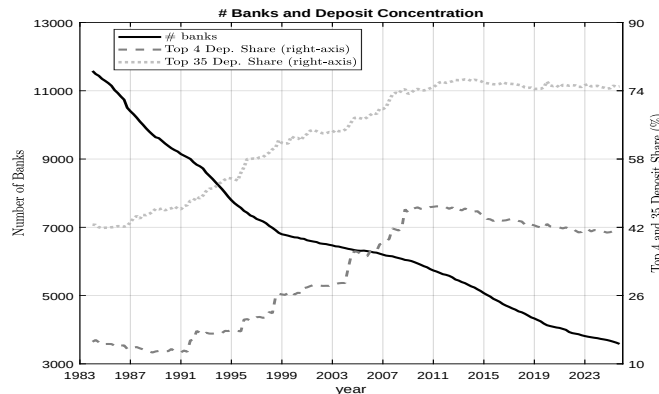
*Written for the Banking and Financial Regulation session at the 2025 World Congress of the Econometric Society. I particularly wish to thank my longtime banking co-authors Pablo D'Erasmus and Ross Levine from whom I have learned so much. I also wish to thank Jason Allen, Will Diamond, Charles Smith, and the research department at the Central Bank of Chile for constructive comments on an earlier version of this draft. The appendix to this paper can be found at Appendix. Some results are still preliminary and hence later versions of this paper can be found on my website at Update.

1 Introduction

Conventional Central Bank monetary policy to maintain price stability and maximum employment works through the provision of liquidity to the banking industry. The effectiveness of monetary policy depends on both the bank lending and deposit channels. Understanding the market structure of the banking industry is important to evaluate the monetary transmission mechanism at the heart of macroeconomic policymaking. Why banks exist at the middle of this intersection of industrial organization and macroeconomics is the subject of the theory of financial intermediation: to perform maturity transformation as in Diamond and Dybvig (1983) fundamental to the deposit channel and delegated monitoring as in Diamond (1984) fundamental to the lending channel.

Financial regulation plays an important role in the market structure of the banking industry and its stability as well as its influence on the bank lending and deposit channels. In fact, founded by an act of Congress in 1913, the Federal Reserve’s primary purpose was to enhance the stability of the American banking system following a series of bank panics. The subsequent widespread bank failures during the Great Depression was a major factor in the Banking Act of 1933, which established deposit insurance. Regulation played a direct role in the industrial organization of the banking sector, starting with the McFadden Act in 1927 which attempted to address long-standing concerns about concentration of financial activity by restricting cross-state bank branching. As Figure 1 makes clear, following the removal of those branching restrictions by the Riegle-Neal Act of 1994, there has been massive consolidation of the banking industry; the top 4 banks now account for approximately 40% of the deposit market share and the top 35 banks (those considered systemically important) comprise over 70% of the deposit market share. In summary, regulatory changes have resulted in much higher concentration with implications for systemic risk.

Figure 1: Concentration Measures and Number of Banks



Notes: Top 4 Deposit Share corresponds to the share of total deposits accounted by the top 4 banks (when sorted by deposits). Top 35 Deposit Share corresponds to the share of total deposits accounted by the top 35 banks. Sample: 1984-2025. Source: Reports of Condition and Income (Call Reports)

Another clear insight from Figure 1 is that there is significant heterogeneity in the bank size distribution. If the nearly 4,000 US banks in 2025 were identical, each would comprise only 0.025% of market share compared to 70% by the systemically important top 35 cohort in the data. A simple characterization of the industry is that there is a group of big, systemically important dominant banks and a competitive fringe. This characterization is in line with the granularity literature of Gabaix (2011) for nonfinancial firms.¹ Applied to the banking sector, granularity implies that an idiosyncratic funding shock (say to deposits) at one of the dominant, top 35 systemically important banks may spill over to the aggregate economy. In contrast, an idiosyncratic shock to even a countable set of small (i.e. measure zero) banks as in competitive models has no aggregate implications.² For example, a pecuniary externality arising from strategic pricing decisions by granular banks arising in response to an idiosyncratic shock may induce negative selection in the loan market that spills over to the aggregate economy through a risk shifting mechanism.³

An important input into understanding the monetary transmission mechanism (as well as Gabaix’s results) is the presence of heterogeneity of a bank’s propensity to lend out of idiosyncratic funding shocks. This ties the banking literature to the heterogeneous agent macroeconomics literature that studies the marginal propensity to consume as surveyed by Kaplan and Violante (2022). In particular, there can be substantial differences between big and small banks in their marginal propensity to lend.

Besides questions of systemic risk, Figure 1 also raises the question about whether the U.S. commercial banking industry has become less competitive and what that might mean for efficiency in the loan and deposit markets. For instance, are markups rising and why? To answer these questions, one approach is to build structural models consistent with data to disentangle these difficult to answer questions (an approach consistent with the Cowles Commission focus on Theory and Measurement). The challenging question then arises as to how to model the industry in Figure 1 and then embed it into a structural framework that is also relevant for monetary and macro-prudential policy making?

The structural framework typically employed by Central Banks is a Dynamic Stochastic General Equilibrium (DSGE) model which assumes a perfectly competitive banking sector with constant returns to scale (CRS) so that bank size is indeterminate, in which case the economy can be characterized by a single representative bank. Important recent examples where banks are subject to financing frictions include Gertler and Karadi (2011) and Gertler and Kiyotaki (2015). These models, however, lack the heterogeneity and potential for idiosyncratic spillovers discussed above.

To that end, this chapter introduces heterogeneity in the banking industry in ways that encompass systemic risk (i.e. where an idiosyncratic shock to a financial institution can

¹Specifically, Gabaix found that idiosyncratic productivity shocks to the top 100 nonfinancial firms accounted for over 30% of GDP volatility.

²A similar logic applies even with banks of positive measure (i.e. atoms) within a continuum of locations or differentiated products since they become atomless in that continuum.

³To fix language in terms of Aumann (1964), in this chapter I will term atomless banks of measure zero as small or fringe and banks of positive measure or atoms as granular or dominant. A more recent set of papers term perfectly competitive economies as “atomistic”; using that definition, Corbae and D’Erasmus (2025) call fringe banks “atomistic” and dominant banks “non-atomistic”.

spillover to the entire financial system through a risk shifting effect) and is consistent with the banking industry data facts described above and to follow in Section 3. One recent approach described in this chapter is to apply the dominant-fringe framework of Gowrisankaran and Holmes (2004) within a quantitative model of the banking industry where banks exist to do delegated monitoring as in Diamond (1984). That model, employed in Corbae and D’Erasmus (2021), can then be used to study the bank lending channel described in the empirical work of Kashyap and Stein (2000) and possible tradeoffs between competition and stability studied by a vast empirical literature as in Beck, Demirguc-Kunt, and Levine (2006) and Berger, Klapper, and Turk-Ariss (2009), as well as the impact of regulatory changes in capital and liquidity coverage ratios. The dominant fringe framework can also be applied to understand the merger wave following the Riegle-Neal Act that contributed to the high concentration of big, dominant banks. Advances in how to compute such quantitative industrial organization models include Ericson and Pakes (1995) and Ifrach and Weintraub (2017).

Another recent quantitative heterogeneous bank approach described in this chapter is Egan, Hortaçsu, and Matvos (2017) where intermediaries exist to do maturity transformation as in Diamond and Dybvig (1983). A shock to beliefs about the stability of one dominant bank can also spillover to other banks even without direct linkages between banks through strategic deposit competition similar to the strategic lending spillover discussed above. Bertrand competition in the pricing of deposits can address the deposit channel of monetary policy as in the empirical paper of Drechsler, Savov, and Schnabl (2017).

While the literature on banking is voluminous, here I will only focus on a subset with apologies to all the excellent research that I cannot find space to review.⁴ I will keep my focus on the role that regulation plays in competition, market efficiency, and stability of the banking industry. My end goal in this chapter is to survey simple quantitative models of the banking industry consistent with data that can be used as a laboratory to conduct relevant policy experiments. What is essential to that task?

- The market structure should be endogenous; regulatory changes (and other exogenous shocks) can change competition.
- The models should be dynamic to capture the charter value of banks and be exposed to solvency and liquidity risk with the potential for spillovers between banks.
- The models should allow for an outside option to address regulatory arbitrage; this insight was made quantitative in Buchak et al. (2018) and addressed in quantitative macro models by Begenau and Landvoigt (2022), Corbae and D’Erasmus (2021), and Dempsey (2025).

The paper is organized as follows. Section 2 provides a brief description of some key U.S. regulatory actions that provide a basis for what our models should address: deposit

⁴There are two insightful handbook chapters that summarize the current literature in macro-finance by Gertler and Kiyotaki (2010) and industrial organization-finance by Clark, Houde, and Kastl (2021). This chapter tries to integrate those into a macro-IO-finance review with a particular focus on the implications of heterogeneity for regulation.

insurance and too-big-to-fail generating moral hazard, governance to address agency frictions, merger policy to address market power, how regulatory arbitrage affects competition within and between banks and nonbanks, among other topics. Section 3 provides a brief description of U.S. banking data facts that provide a basis for model selection and moment matching/validation of our models. Section 4 provides a brief description of the economic environments of some of the fundamental banking theories that underlie the quantitative models presented in the remaining sections. Section 5 provides a very simple dynamic model to frame the issues alluded to in the title of this chapter: competition, efficiency, and stability. The models of Sections 4 and 5 exploit symmetry to maintain tractability. Given the substantial heterogeneity in the data described in Section 3, the remainder of the paper describes structural models with heterogeneity that can be used for policy analysis including big, granular banks. Section 6 presents a dynamic quantitative model that focuses on the deposit side of the bank balance sheet. Section 7 presents a dynamic quantitative framework to understand the merger wave following the removal of cross-state branching restrictions in the Riegle-Neal Act. Section 8 presents a quantitative dynamic framework that focuses on the loan side of the bank balance sheet to understand the bank lending channel of monetary policy and regulatory arbitrage. Finally, Section 9 provides directions for future research. An Online Appendix in Section A contains numerous tables, graphs, and further content that are referenced in the chapter.

2 A Little Financial Regulation History

Good regulatory banking policy is designed to remedy certain underlying financial frictions, otherwise it may create a friction itself. Those policies may also face tradeoffs. For example, bank deposit insurance to prevent non-fundamental runs may create a moral hazard problem inducing excessive risk taking. Capital requirements to correct excessive risk taking by imposing more “skin-in-the-game” by equity holders may induce regulatory capital arbitrage whereby banks take their risky loans off-balance-sheet and/or shift lending to unregulated (shadow) nonbanks studied by Buchak et al. (2024).

This section will present a short background on key regulatory acts often in response to crises in order to shed light on some of the underlying frictions that the following models should incorporate. I will focus on a subset of Acts in chronological order relevant for the models I will present.

The Federal Reserve Act was signed into law by President Woodrow Wilson on December 23, 1913 in response to earlier financial panics which highlighted the need for a central bank (which was absent in the U.S. up to that time) to provide stability to the American financial system. Such panics were widely blamed on the nation’s “inelastic currency.” Specifically, the supply of notes issued by national banks was largely unresponsive to changes in demand, especially when news caused bank customers to run on their bank to withdraw cash. To respond to this problem, the Act established the Federal Reserve Note as the national currency in response to the lack of an elastic funds in response to liquidity needs of banks. For this chapter’s purposes, one can interpret this as a policy response to bank specific liquidity shocks with incomplete insurance markets. One equilibrium response to those

idiosyncratic shocks is for a bank to grow in order to diversify its deposit base (something we will see in the data Section 3 and take up in the model Section 7). Idiosyncratic deposit risk also provides a rationale for government provision of liquidity at the Fed’s discount window (which began in 1913) and an interbank market (e.g. Fed Funds which began in the 1920s). In this chapter, I will not have space to address models of the interbank market but instead point the reader to Afonso and Lagos (2015) and Chapter 5 of Hugonnier, Lester, and Weill (2025).

The Banking Act of 1933 (also known as Glass-Steagall) aimed to stabilize the US banking system after the Great Depression. Among other things, the 1933 Act created the Federal Deposit Insurance Corporation (FDIC). While the creation of deposit insurance helped stabilize pessimistic beliefs and end bank runs (the focus of Diamond and Dybvig (1983)), it also created a moral hazard problem in that banks did not face the external funding cost of excessive risk taking (the focus of Keeley (1990)). The 1935 Banking Act made deposit insurance permanent. Thus, our models should address moral hazard and excessive risk taking along with the potential for belief based runs, some of which brought about later regulatory actions (like the Dodd-Frank Act).

The Glass-Steagall Act also separated commercial and investment banking activities in order to prevent banks from using depositors’ money for risky investments like stock speculation. The Glass-Steagall Act was later replaced by the Financial Services Modernization Act of 1999 (also known as the Gramm–Leach–Bliley Act (GLBA)) which removed barriers in the market among banking companies, securities companies, and insurance companies that prohibited any one institution from acting as any combination of an investment bank, a commercial bank, and an insurance company. With the passage of GLBA, commercial banks, investment banks, securities firms, and insurance companies were allowed to *consolidate*. From an industrial organization perspective, GLBA can be thought of as providing holding companies the opportunity to exploit *economies of scope*.

As discussed in the introduction, the Riegle-Neal Interstate Banking and Branching Efficiency Act of 1994 removed many of the restrictions on opening bank branches across state lines imposed under the McFadden Act of 1927. The ensuing merger wave, as big banks bought up smaller ones across state lines, increased dominant bank market shares until the “Great Recession” (running from December 2007 to June 2009).⁵ The DOJ evaluates a bank merger’s competitive impact, while the banking regulators evaluate the merger based on five key criteria including macro risks to the U.S. banking or financial system (i.e. financial stability). As will be discussed in Section 3, Riegle-Neal allowed banks to grow their deposit base in order to exploit *economies of scale* and diversify idiosyncratic shocks across regions. Quantitative models that can assess how endogenous changes in concentration affect market efficiency as measured by markups and interest margins along with its implications for financial stability will be presented in Sections 7 and 8 below.

There are numerous regulatory acts that have attempted to address corporate gover-

⁵Bank mergers are subject not only to the same federal antitrust laws as other sectors, but also to an industry-specific regulatory review framework by the Department of Justice (DOJ) and at least two federal banking agencies (Office of Comptroller of the Currency, FCC, Fed, and in many cases, by state authorities as well). A related question of whether there is tacit collusion in the banking industry is taken up in Corbae and Gofman (2023).

nance frictions. For instance, Section 112 of the FDIC Improvement Act of 1991 extended the independent-audit-committee requirement to every insured institution with greater than \$500 million in assets and those above \$3 billion also had to place at least two directors with “banking or financial management expertise” on that committee. Numerous other later regulatory acts (e.g. Sarbanes-Oxley Act of 2002, Dodd-Frank Act of 2010, and OCC “Heightened Standards” for Large Banks in 2014) tightened board-level oversight of managerial compensation and risk taking. To assess governance issues associated with a potential conflict between bank managers and owners over risk taking, Laeven and Levine (2009) conducted the first empirical assessment of the relation between risk taking, ownership structure, and bank regulations. They show that ignoring ownership structure leads to incomplete and sometimes erroneous conclusions about the impact of regulatory policy. Their empirical work motivates us to include potential agency conflicts between owners and managers in our models, highlighted in Section 5 below.⁶

Finally, the financial crisis behind the Great Recession prompted much regulatory action in the Dodd-Frank Wall Street Reform and Consumer Protection Act of 2010. The significant regulatory changes along with Basel III included tighter capital and liquidity requirements that vary with bank size and the state of the economy, stress testing of systemically important banks whose distress can spill over to the rest of the financial system, orderly resolution of failing banks as a policy solution to the too-big-to-fail (TBTF) moral hazard problem, and other provisions like the Volcker Rule which restricted banks from making certain kinds of speculative investments. The fact that many of these regulatory changes were bank size dependent illustrates why models with bank heterogeneity provide a potentially important laboratory to conduct macro-prudential policy experiments.⁷ Quantitative models with bank heterogeneity will be presented in Sections 6-8.

3 Some Banking Data Facts to Guide Model Selection

Here we will simply list some of the key banking data facts that motivate the models we focus on in later sections. These facts are explained in more detail in Corbae and D’Erasmus (2021), Corbae and D’Erasmus (2025), and Corbae, D’Erasmus, and Smith (2025). As in Kashyap and Stein (2000), the main data source for this section is the Consolidated Report of Condition and Income (known as Call Reports) that banks submit to the Federal Reserve each quarter.⁸

⁶Chen et al. (2022) use a difference-in-difference approach associated with that they state (p.146) as “...perhaps the most significant shock to transparency that occurred in a long time: The Sarbanes-Oxley Act” to test whether heightened disclosure regulation inducing more transparency has significant impact on depositor’s behavior.

⁷A simple linear factor model to conduct stress tests is developed in Begenau, Piazzesi, and Schneider (2025).

⁸The number of institutions and its evolution over time can be found here. Balance Sheet and Income Statements items can be found here. In the Data Appendix of Corbae and D’Erasmus (2025), we provide a detailed description of variable definitions and how all the statistics reported are constructed. We also use the Summary of Deposits (SOD), the annual survey of branch office deposits as of June 30 for all FDIC-insured institutions to be found here.

In Corbae and D’Erasmus (2025), we provide evidence for:

1. Imperfect competition:
 - (a) top 4 and 35 banking industry concentration as in Figure 1 above,
 - (b) incomplete passthrough of cost to revenue as measured by the Rosse-Panzer H-statistic in Figure 3 rejecting the perfect competition level of 1.
 - (c) rising markups as concentration increases and higher markups for top 35 banks than fringe in Figure 6 (see also Figure 2 of Corbae and D’Erasmus (2021)).⁹
2. Large right tail deviations from Zipf’s law in deposits in Figure 4 (with statistically significant differences from the Power law prediction estimated in Table A.2).
3. Geographic diversification by the top 4 and 35 banks in Figure 5. Notably, the number of states with active branches by the (top 4, top 5-35, outside top 35) grows from (3.5,3.4,1.04) in 1994 to (35.0,10.5,1.3) in 2023.
4. Diversification of idiosyncratic deposit shocks resulting in lower variance of funding inflows in Table 1. Specifically, estimates from a dynamic panel yield the variance of deposits by bank size (top 4, top 5-35, outside top 35)=(0.118,0.247,0.362). Variation in deposit inflows can explain why banks of different sizes hold different buffer stocks of capital in excess of capital requirements documented in Figure 3 of Corbae and D’Erasmus (2021).
5. Increasing returns consistent with banks’ cost structure in Table 2. Specifically, asset weighted average costs (sum of average marginal net expense and average fixed cost scaled by loans) are decreasing in size (top 4, top 5-35, outside top 35)=(1.47,1.77,2.62). Our findings are consistent with economies of scale and scope as well as previous empirical results as in Berger and Mester (1997).
6. Countercyclical failure rates in Figure 7 as well as the predominance of bank entry and exit by small banks in Table A.3.
7. Consistent with a diversification argument for big banks, the standard deviation of charge-off rates, default frequencies, loan returns, net interest margins, loan interest rates, and loan interest spreads decline with bank size in Table A.4.
8. While we do not find statistically significant differences in means across banks of different sizes in the cost of deposits, loan interest rates, or the return of (net) securities, we do find statistically significant differences in markups and the Lerner index increasing with bank size in Table A.4.¹⁰

⁹See other recent evidence for rising markups in the banking industry by Benkard, Miller, and Yurukoglu (2025).

¹⁰The fact that there are not statistically significant differences in deposit costs, loan interest rates, and securities returns factors into certain modeling assumptions in Sections 7 and 8.

9. Countercyclical default frequencies, chargeoffs, net interest margins, and markups in Table A.5. The table also documents we do not find evidence of statistically significant cyclical movements in the cost of deposits, loan returns, or net securities returns.

In Corbae, D’Erasmus, and Smith (2025), we provide empirical evidence:

- i. Establishing that the majority of mergers are by big (dominant) banks acquiring small (fringe) banks in Figure 5. Specifically, we document that 96% of bank mergers involve an acquirer that is significantly larger than its target and the median acquirer has over seven times higher market capitalization than the target. Further, we show that the mean merger price of the target is only 1.1 times the target’s market value.
- ii. In Figure 6, we show that the number of mergers declined since the peak after the introduction of Riegle-Neal to the financial crisis. It has rebounded slightly but generally remained below pre-crisis levels. In Figure 3, we demonstrate that merger rates are procyclical in Figure 3. Specifically, the correlation between the merger rate and detrended GDP between 1984-2023 was 0.167 while the correlation in the “transition path” following Riegle-Neal between 1994-2019 was 0.354.
- iii. For average county level bank HHI’s are 2191 over 1984-1993 and 2606 from 2010-2019 in Table A.3. Table 7 of Nocke and Whinston (2022) documents that the majority of non-financial anti-trust cases that fall in that range are enforced.¹¹
- v. Despite rising Herfindahl indices suggesting lower competition *within* the banking sector, the growing non-bank sector provides an outside borrowing option that competes with banks *across* the credit market. Specifically, we document that Total Credit to the Private Non-Financial Sector by Banks to Total Credit to the Private Non-Financial Sector has fallen from 45% from 1984-1993 to 33% from 2010-2019 in Table 4.
- vi. For heterogeneity in the marginal propensity to lend (MPL) out of idiosyncratic innovations to deposits between large and small banks in Table 1. Specifically, the top 4 banks have an average MPL of 0.959 while all other banks have an average MPL of 0.510. We also show that the MPL for top 4 banks declines significantly over time.
- vii. For a granular view (in the language of Gabaix (2011)) of how idiosyncratic shocks to bank deposit flows of the largest U.S. banks spill over to the aggregate economy via changes in bank credit in Table 2 as well as total credit and gdp in Tables A.7 and A.8. Specifically, we document that idiosyncratic shocks to top 4 bank deposits can explain 10% of aggregate bank lending growth.¹²

¹¹A merger is “enforced” if the FTC sought to block or modify it. The categories for the FTC report are: grocery, oil, chemical, pharmaceutical, hospital, electronically-controlled devices and systems, branded food/beverages, and other

¹²While we focus on idiosyncratic shocks to bank funding, Galaasen et al. (2023) provide evidence that idiosyncratic shocks to large borrowers is not diversified away on bank loan portfolios effectively creating “aggregate” credit risk for large banks.

- viii. Figure 4 documents an increase in the ratio of banks’ market value to equity (i.e. Tobin’s Q) following Riegle-Neal until the financial crisis using market value to book equity as in Begenau et al. (2026). This is consistent with “profitable reallocation opportunities” studied in Jovanovic and Rousseau (2002) and our findings of greater allocative efficiency in Table 9 following the relaxation of regulation.

All models must simplify the economic environment they are intended to represent. Table 1 in the Online Appendix provides a simple representation of bank balance sheets in several of the models considered below by asset size for the data used in Figure 1. The liability side includes deposits (the share of which declines with higher asset size), fed funds borrowed (rising share), and equity (declining share). The asset side is loans (the share of which rises with higher asset size), securities (declining share), and fixed/intangible assets (declining share).

These data facts suggest modeling banks along the lines of the Diamond (1984) delegated monitoring framework capturing increasing returns and geographic diversification and potential shocks to deposits along the lines of Diamond and Dybvig (1983). Motivated by rising levels of concentration and markups, a model with imperfect competition along the lines Allen and Gale (2004) and Boyd and De Nicolo (2005) may be important. High concentration of the top 35 banks versus the other 4000 fringe banks, may best be modeled along the dominant-fringe framework Gowrisankaran and Holmes (2004). That framework can also be used to address the transition between the relatively stable low level of concentration before Riegle-Neal and the relatively stable high level of concentration following the Great Recession evident in Figure 1 as dominant banks make take-it-or-leave-it offers to fringe banks in order to grow and diversify their deposit base. The fact that there has been rising loan market competition from non-banks, suggests the models should account for a non-bank (or shadow) sector along the lines of Buchak et al. (2018) to help capture regulatory arbitrage. The fact that charge-off rates and default are countercyclical as is bank failure suggests quantitative models should allow for endogenous exit (and entry) generating a size distribution of banks consistent with Figure 1. We turn now to some of the fundamental applied theory papers cited above.

4 Building Block Theory

As the previous section suggests, there are several important papers that can provide the building blocks for dynamic quantitative models of the banking industry. In order to provide a self-contained chapter on banking and financial regulation, in this section I briefly describe some key papers that are at the foundation of the quantitative models in later sections.

4.1 Maturity Transformation

Starting with the liability side of a bank balance sheet, banks exist in the Diamond and Dybvig (1983) model to provide maturity transformation by backing short-term liabilities (that are payable on demand to households (hhs) with random liquidity needs) with produc-

tive long-term illiquid assets.¹³ A constrained efficient arrangement is to pool resources and provide insurance against the privately observed idiosyncratic liquidity shocks. The efficient allocation can be decentralized as a bank that offers demand deposits. The problem is that bank deposits are susceptible to multiple equilibria: (1) the efficient allocation where only enough liquidity is withdrawn to meet fundamental needs and (2) an inefficient run where too much liquidity is withdrawn from the productive technology so that the bank becomes insolvent.

To put some structure on the model, there are three periods $t = 0, 1, 2$ and a unit measure of ex-ante identical hhs. All hhs have 1 unit of a good at $t = 0$ but prefer to consume either at $t = 1$ or $t = 2$. There are two storage technologies. A productive long term technology: 1 unit of goods invested at $t = 0$ yields $R > 1$ units at $t = 2$ but yields 1 if the storage is interrupted at $t = 1$ (i.e. the salvage value is the initial investment). There is also a technology such that goods at $t = 1$ can be stored one-for-one (i.e. under a pillow) until $t = 2$. While ex-ante identical, each hh receives an unobservable shock at $t = 1$ determining their preferences: with probability π they are “early” hhs who only want to consume at $t = 1$ with preferences $u(c_1)$ and with probability $(1 - \pi)$ they are “late” hhs who only want to consume at $t = 2$ with preferences $u(c_2)$ where $u(c_t)$ exhibits risk aversion. While hhs face risk, given the shocks are i.i.d. across the unit measure of hhs, there is no aggregate uncertainty so that hh risk can be perfectly pooled and π is the fraction of early hhs.

4.1.1 An efficient arrangement

An efficient arrangement solves

$$\max_{(c_1 \geq 0, c_2 \geq 0)(S, W) \in [0, 1]} \pi u(c_1) + (1 - \pi)u(c_2) \quad (1)$$

$$\text{subject to } S + \pi c_1 = W \quad (2)$$

$$(1 - \pi)c_2 = R(1 - W) + S \quad (3)$$

$$u(c_1) \geq u(0) \quad (4)$$

$$u(c_2) \geq u(c_1) \quad (5)$$

where (1) is the utilitarian (equal weighted) objective, (2)-(3) are resource constraints at $t = 1$ and $t = 2$ respectively, and (4)-(5) are incentive compatibility constraints ensuring truthtelling by early and late hhs.

Since there is no aggregate risk and $R > 1$, it is strictly better not to liquidate more than necessary to cover expenditure by early hhs, so $S = 0$. Neglecting the incentive compatibility constraints, the optimal allocation implies

$$u'(c_1^*) = Ru'(c_2^*) \implies c_1^* < c_2^* \quad (6)$$

since $R > 1$ and $u'(c_t) > 0$. But $c_1^* < c_2^*$ implies the incentive compatibility constraints are non-binding and after substituting W^* from (2) into (3) yielding $\pi c_1^* + (1 - \pi)c_2^*/R = 1$ and (6) is 2 equations in 2 unknowns to solve for the efficient allocation (c_1^*, c_2^*) .

¹³In Diamond (2020), financial intermediaries exist to meet households’ demand to hold risk free assets and create value even when holding liquid securities. There, intermediaries perform safety transformation: buying a pool of liquid risky assets to back the issuance of a senior debt tranche that is risk-free.

Given risk aversion, an efficient arrangement smooths consumption relative to autarky where $(c_1^A, c_2^A) = (1, R)$; i.e. $1 < c_1^* < c_2^* < R$. The efficient allocation improves on autarky by transferring resources from a fraction $1 - \pi$ of late hhs who receive R to the fraction π of early hhs who only receive 1. Notably, it is not resource feasible to implement the efficient allocation if the *entire* unit measure of hhs were to consume c_1^* ; it is this insight which explains why a decentralization of the efficient allocation via banks may be prone to runs.

4.1.2 Decentralization

One way to decentralize this allocation is to create a Diamond and Dybvig (1983) competitive bank which offers a deposit contract to a hh depositing their one good at $t = 0$ and offering a return $r_1 = c_1^*$ if it withdraws at $t = 1$ and $r_2 = c_2^*$ if it withdraws at $t = 2$. This contract is only resource feasible if π hhs withdraw at $t = 1$ and $(1 - \pi)$ withdraw at $t = 2$. As suggested above, if all households (both early and late) withdraw at $t = 1$, this contract is resource infeasible (i.e. the bank would be insolvent). This insolvency can arise if atomless (measure zero) hhs believe that all other hhs will withdraw at $t = 1$. Under that belief, it is individually rational for a late hh to withdraw at $t = 1$ since there is a chance it will receive $r_1 = c_1^*$ whereas if it waits and attempts to withdraw at $t = 2$, it receives nothing. That is, it is rational to run under a pessimistic set of beliefs that the bank has a solvency problem.

Not only does the paper make an important contribution to understanding the existence of nonfundamental bank runs, but it also provides a rationale for deposit insurance. Specifically, if the government (i.e. FDIC) commits to insure depositors such that even if the bank is not able to fulfill its obligations, the depositors receive the full value of their deposits, then a late consumer has no incentive to run and deposit insurance implements the first best without any taxes actually ever being paid. One potential problem, however, is the credibility or time consistency of the commitment to deposit insurance. For instance, when deposit insurance actually has to be paid out, might the government choose not to honor that obligation? This important question is taken up in Ennis and Keister (2009).

In summary, besides providing a rationale for why banks exist (to do maturity transformation), this is a very important and powerful insight from theory that can help us understand the data and regulatory responses to the data. Bank runs are not only a thing of the past as the recent experience of Silicon Valley Bank exposed. One critical question that always remains is what is the trigger? Is it fundamental or non-fundamental?¹⁴

4.2 Delegated Monitoring

Moving on to the asset side of a bank balance sheet, banks exist in the Diamond (1984) model to provide loan monitoring services in the presence of private information on the part of borrowers. Specifically, suppose borrowers have private information about the success or failure of their idiosyncratic project which can only be made observable to the lender at a

¹⁴One recent paper that has taken up this question is Correia, Luck, and Verner (2025).

fixed cost.¹⁵ If the project is not monitored then the borrower can claim it failed and run away with the proceeds. Further, suppose the borrower's financing needs are large but private investors (i.e. depositors) have small amounts of funds so that it takes many of them to fund the project. Then instead of having each investor pay a fixed cost to monitor, an efficient financial arrangement would delegate monitoring to one agent (the bank) to economize on those costs. The problem with this arrangement, however, is that the monitor has all the information and can also potentially claim failure and run away with the proceeds. In that case, who is going to monitor the monitor? The fundamental result in Diamond's paper is that if the bank is sufficiently large (i.e. has funded many idiosyncratic projects on its asset balance sheet and hence is sufficiently diversified), then it cannot claim insolvency (and run away with the proceeds).

To put some structure on this static model, suppose there are N identical risk neutral entrepreneurs/firms that each want to borrow at the beginning of the period to finance an indivisible productive project requiring 1 unit of capital which pays off at the end of the period. Firm n 's project yields a random amount $\tilde{Y}_n \in [0, \bar{Y}]$ which is observable only to the firm. These returns are identically and independently distributed (i.i.d.) across firms. By paying a monitoring cost K , however, a lender can observe $\tilde{Y}_n$. Assume that monitoring costs are not too large so that lending is profitable for investors relative to investing in a riskless technology with gross return R . That is, $E[\tilde{Y}_n] - K > R$ for each n . Risk neutral private investors are small; each has only $1/m$ units of capital (so m of them are necessary to finance a project) where $m > 1$ is assumed to be big. So that all projects can possibly be financed, assume there are at least $m \cdot N$ private investors. There is also an auditing technology such that if an agent (be it a borrower or even a bank) declares it is insolvent (i.e. cannot pay what it promised), then it bears a bankruptcy penalty C .

In this environment it is simple to see why it is cost efficient to delegate the monitoring to one risk neutral agent (i.e. a bank) instead of an arrangement where each small investor directly invests in a firm. Specifically, the total monitoring cost from lending to N profitable firms in this economy from direct finance is $N \cdot m \cdot K$ while monitoring costs from bank finance would only be $N \cdot K$.

But this calculation does not include the implicit costs associated with delegating monitoring to the bank. That is, what is the incentive for the bank to report truthfully in this static model? Instead, it could claim that all the firms failed and abscond with the difference (i.e. $\left[\sum_{n=1}^N \tilde{y}_n\right] \geq 0$) where $\tilde{y}_n$ is the payment from firm n to the bank with feasibility requiring $\tilde{y}_n \leq \tilde{Y}_n$.

So, how to incentivize a bank to report truthfully (i.e. monitor the monitor)? Obviously, total payments to the bank depends on the observable number of loans N on the bank's balance sheet. Diamond assumes the bank offers a noncontingent debt or "deposit" contract under which each investor is promised R_D^N/m in exchange for a deposit of size $1/m$ at the beginning of the period and the bank is audited/liquidated if its announced cash flow at the end of the period net of monitoring costs from its portfolio of N loans, $\tilde{z}^N$, is less than the

¹⁵Other important papers along this dimension are Townsend (1979) and Williamson (1986), as well as the clear exposition in Freixas and Rochet (2008).

total sum promised to depositors, $N \cdot R_D^N$. This contract can be made incentive compatible through the threat of liquidation imposing costs C^N on the bank. The equilibrium deposit rate R_D^N and audit penalties C^N satisfy:

$$\mathbb{E} \left[\min \left\{ \frac{1}{N} \sum_{i=1}^N \tilde{y}_i - K, R_D^N \right\} \right] = R \quad (7)$$

and

$$C^N = N \cdot \mathbb{E} [\max \{ R_D^N - \tilde{z}^N, 0 \}]. \quad (8)$$

Given the outside return option of R , equation (7) pins down R_D^N given bank solvency and truth-telling induced by (8). Specifically, incentive compatibility is ensured by (8) since if the bank announces $\tilde{z}^N < \frac{1}{N} \sum_{n=1}^N \tilde{y}_n - K$, it is audited and penalized

$$R_D^N - \left(\frac{1}{N} \sum_{n=1}^N \tilde{y}_n - K \right)$$

in which case it has no incentive to lie. Notice that a truthful announcement $\tilde{z}^N = \frac{1}{N} \sum_{n=1}^N \tilde{y}_n - K$ by the bank bears no cost in (8) given it is not insolvent by (7).

Diamond proves the following proposition providing a rationale for why banks exist to deliver efficient monitoring.

Proposition 1 (Diamond (1984) Prop. 2). *If monitoring is efficient ($K < C^1$), investors are small ($m > 1$), and investment is profitable ($\mathbb{E}[\tilde{Y}] - K > R$), financial intermediation (delegated monitoring) dominates direct lending if N is large enough (i.e. the bank is sufficiently diversified).*

Proof. Bank finance dominates direct lending if

$$K + \frac{C^N}{N} < m \cdot K. \quad (9)$$

It is sufficient to show that $\frac{C^N}{N} \rightarrow 0$ as $N \rightarrow \infty$. The strong law of large numbers implies that

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N \tilde{y}_n = \mathbb{E}[\tilde{y}].$$

Then since $\mathbb{E}[\tilde{Y}] - K > R$, (7) implies $\lim_{N \rightarrow \infty} R_D^N = R$ (i.e. deposits are riskless). Finally, (8) implies $\lim_{N \rightarrow \infty} C_N/N = 0$. $\square$

In summary, this is a very important and powerful insight from theory that can help us understand the data; with scale economies in monitoring, diversification helps lower the costs of providing incentives for good behavior (i.e. truth-telling) by bank managers.

4.3 Competition and Risk Taking

Neither of the two previous papers providing a foundation for why banks exist explicitly address the industrial organization of the banking industry in Figure 1 and the role of competition in risk taking. In an important paper, Keeley (1990) pointed out that a fixed-rate deposit insurance system provides a moral hazard for excessive risk taking and is not viable absent regulation. Although the deposit insurance system appeared to have worked remarkably well over most of its 50-year history, major problems began to appear in the early 1980's. His paper tests the hypothesis that increases in competition caused bank charter values to decline, which in turn caused banks to increase default risk through increases in asset risk and reductions in bank capital.

Allen and Gale (2004) and Boyd and De Nicolo (2005) provide a simple framework to attempt to understand the relation between bank concentration and financial stability, which is the focus of an extensive empirical literature. For example, using a panel of 69 countries and controlling for many factors such as regulatory differences, Beck, Demirguc-Kunt, and Levine (2006) find banking concentration (market share of top 1%) is negatively related to the probability of a banking crisis (e.g. 2x higher exit rate), a result which is sometimes called the “competition-fragility” hypothesis. On the other hand, Berger, Klapper, and Turk-Ariss (2009) find that concentration is positively related to default frequency, a result which is sometimes called the “competition-stability” hypothesis.

4.3.1 On Competition-Fragility

To put some structure on the static (i.e. beginning and end of period) model of Allen and Gale (2004), there are two sets of risk neutral agents: bankers and depositors. The N bank(er)s have no initial resources but have access to a set of constant return-to-scale risky technologies indexed by R . The risky technology yields $R \in [0, \bar{R}]$ with probability $p(R)$ and 0 otherwise where $p(0) = 1$, $p(\bar{R}) = 0$, $p' < 0$, $p'' \leq 0$. That is, choosing a riskier technology lowers the likelihood of success.¹⁶ (R, p) defines a risk-return tradeoff of the bank's loan portfolio. $p(R) \cdot R$ is a strictly concave function of R and reaches a maximum R^* when $p'(R^*)R^* + p(R^*) = 0$. Increasing R from the left of R^* entails increases in both the probability of failure and expected output. The bank's choice of R at the beginning-of-period is unobservable to outsiders. At the end-of-period, outsiders can only observe and verify at no cost whether the investment's outcome has been successful (positive output) or unsuccessful (zero output). By assumption, contracts are simple debt contracts. The bank has complete control over the choice of risk taking R .

The supply of deposits is represented by an upward sloping (convex) inverse supply curve, denoted by $r_D(\cdot)$. That is, to obtain more funding, a bank needs to raise its deposit rate.

¹⁶The assumption that each bank chooses a portfolio consisting of perfectly correlated risks is equivalent to assuming that the risk of each investment can be decomposed into a common component and a purely idiosyncratic component. If there is a very large number of investments, the purely idiosyncratic components can be pooled perfectly (as in Diamond (1984)). Then the idiosyncratic risks disappear from the analysis and we are left with a common component representing the systematic risk modeled above. See Martinez-Miera and Repullo (2010) for more general processes that yield interesting non-monotonicities in the possible tradeoff between competition and stability.

Deposits of bank i are denoted by D_i and total deposits by $\sum_{i=1}^N D_i$. Deposits are insured, so that the relevant supply does not depend on risk and banks pay a flat rate deposit insurance premium $\alpha > 0$.

Banks compete for deposits in a Cournot-Nash game. The deposit interest rate is a function of total deposits: $r_D = r_D\left(\sum_{i=1}^N D_i\right)$. In a Nash equilibrium, each bank i chooses (R_i, D_i) as the best response to the strategies of other banks subject to limited liability to maximize expected profits

$$p(R_i) \cdot \left(R_i - r_D \left(\sum_{i=1}^N D_i \right) - \alpha \right) \cdot D_i + (1 - p(R_i)) \cdot 0 \quad (10)$$

In a symmetric interior equilibrium where $(R_i, D_i) = (R, D) > 0$ and $p(R) > 0$, the first order necessary conditions with respect to (R_i, D_i) reduce to

$$R_i : p'(R) \cdot (R - r_D(N \cdot D) - \alpha) + p(R) = 0, \quad (11)$$

$$D_i : p(R) \cdot (R - r_D(N \cdot D) - \alpha) - p(R) \cdot r'_D(N \cdot D) \cdot D = 0. \quad (12)$$

Allen and Gale prove that these two equations in two unknowns have a unique solution. They also prove the following proposition:

Proposition 2 (Allen and Gale (2004) Prop.5). *In a symmetric equilibrium, the equilibrium level of risk shifting R is strictly increasing in N . As $N \rightarrow \infty$, $R \rightarrow \bar{R}$.*

Simply, the proposition states that more deposit market competition raises the cost of attracting funds (thereby lowering profits) which leads to more risk taking given limited liability. This provides a theoretical foundation for the “competition-fragility” view consistent with the data in, for example, Beck, Demirguc-Kunt, and Levine (2006).

4.3.2 On Competition-Stability

As in Boyd and De Nicolo (2005), now suppose there are many identical entrepreneurs, who have access to projects of fixed size, normalized to 1, with the two-point random return structure previously described. The entrepreneurs have no internal funds, so they need to borrow from banks. Banks cannot observe their risk shifting choice R , but take into account the best response of entrepreneurs’ loan demand to their choice of the loan rate r_L .

Specifically, the demand for a unit loan to fund the entrepreneurs project must solve the following problem. Given a loan rate r_L , each entrepreneur chooses $R \in [0, \bar{R}]$ subject to limited liability to maximize their expected profitability

$$p(R) \cdot (R - r_L) + (1 - p(R)) \cdot 0.$$

The first order necessary condition with respect to R satisfies

$$p(R) + p'(R) \cdot (R - r_L) = 0 \iff h(R) \equiv R + \frac{p(R)}{p'(R)} = r_L. \quad (13)$$

It is simple to show that an increase in loan rates causes entrepreneurs to choose more risk (i.e. $\frac{dR}{dr_L} > 0$).¹⁷ This is the key channel that can reverse the prior result that more

¹⁷Taking the total differential $\frac{dR}{dr_L} = \frac{1}{h'(R)} > 0$ since $h'(R) = 1 + \frac{p'(R)^2 - p(R)p''(R)}{p'(R)^2} > 0$ given $p' < 0, p'' \leq 0$.

competition leads to more fragility; specifically, if more competition lowers loan rates, then borrowers take less risk. One can interpret this as a selection effect; higher interest rates induce only risky borrowers to take loans.

Assume the inverse demand for loans $r_L(L)$ is a decreasing function of the total supply of loans L where $r'_L < 0, r''_L \leq 0$ with $r_L(0) > r_D(0)$. Since banks only fund loans via insured deposits, $L = \sum_{i=1}^N D_i$. Then bank i chooses D_i to maximize

$$p(R) \cdot \left(r_L \left(\sum_{i=1}^N D_i \right) - r_D \left(\sum_{i=1}^N D_i \right) - \alpha \right) \cdot D_i \quad (14)$$

subject to (13). Since $R = h^{-1} \left(r_L \left(\sum_{i=1}^N D_i \right) \right)$ by (13), the bank internalizes that its choice of D_i affects the likelihood of success $p(R)$ of its loan portfolio.

Boyd and DeNicolò prove the following proposition:

Proposition 3 (Boyd and De Nicolò (2005) Prop. 2). *In a symmetric Cournot Nash Equilibrium, the equilibrium level of risk shifting R is strictly decreasing in N . As $N \rightarrow \infty$, the Nash equilibrium converges to the competitive outcome $r_L = r_D + \alpha$.*

Simply, as banks obtain increased market power in the loan market, they use it to raise loan rates, *ceteris paribus*. But then, borrowers optimally respond by increasing the risk of their investment projects. This provides a theoretical foundation for the “competition-stability” view consistent with the data in, for example, Berger, Klapper, and Turk-Ariss (2009).

In related work, Martínez-Miera and Repullo (2017) show that safer entrepreneurs will be funded by nonmonitoring banks and riskier entrepreneurs by monitoring banks. Then, in response to an increase in savings (e.g. the “Savings Glut”) which reduces interest rates and spreads, the relative size of the nonmonitoring banking system increases along with the probability of failure of monitoring banks. Their model rationalizes the existence of counter-cyclical risk premia and the connection between low interest rates, tight credit spreads, and the buildup of risks during booms consistent with events leading up to the financial crisis of 2008.

In summary, these are very important and powerful insights from theory that can help us understand the data; changes in market structure can affect financial stability, hence the need for regulatory oversight by financial authorities like the OCC, FDIC, Fed, and DOJ.

4.4 Spatial Models and Industry Structure

As Data Fact 3 in Section 3 documents, following the Riegle-Neal Act in 1994 there was massive geographic expansion of banks across state boundaries. With idiosyncratic shocks across states, geographic expansion provides diversification for bank assets and liabilities as in Fact 4 in Section 3. Much recent banking literature exploits spatial variation in deposit markets (see in particular, Aguirregabiria, Clark, and Wang (2025), d’Avernas et al. (2024), Morelli, Moretti, and Venkateswaran (2025), and Oberfield et al. (2024)).

In the previous sections, deposits and loans were homogeneous goods. Here we consider the simplest possible spatial model of market structure along the lines of Salop (1979) to

illustrate why a decentralized market may be inefficient and require government regulation.¹⁸ Further, in the previous section, the number of banks N was exogenous while here the number of banks will be determined endogenously, something I will carry over to many of the quantitative models presented in later sections.

To put some structure on this static model, assume a unit measure of identical households (hhs) are uniformly distributed on a unit circle. Banks are indexed by $n = 1, 2, \dots, N$ located on the circle, but because there is a fixed cost F to set up a bank, there will only be a finite number of them. Assume that households can only save in a bank, but there is a cost γx for a hh who lives at a location x distance from a bank to travel to the bank. Assume $\gamma > F$.

4.4.1 The Efficient Number of Banks

Depending on the number of banks N , since households are uniformly distributed on the line, the optimal (in terms of minimizing travel costs) placement of banks are at locations $\{\frac{1}{2N}, \frac{3}{2N}, \frac{5}{2N}, \dots, \frac{2n-1}{2N}, \dots, \frac{2N-1}{2N}\}$. For an economy with N banks, the total cost for hhs to travel to banks is $\frac{\gamma}{4 \cdot N}$.¹⁹ If each bank costs F to set up, then the total fixed cost of setting up the N banks is $F \cdot N$. Hence, the total costs TC to society of this arrangement is given by the convex function $TC(N) = \frac{\gamma}{4 \cdot N} + F \cdot N$.

In that case, the least cost efficient number of banks is given by the first order necessary and sufficient condition with respect to N or

$$-\frac{\gamma}{4N^2} + F = 0 \iff N^* = \frac{1}{2} \sqrt{\frac{\gamma}{F}}. \quad (15)$$

Thus, the socially optimal number of banks is decreasing in the fixed cost of setting up a bank and increasing in travel costs.

4.4.2 Decentralized Arrangement

Now consider a decentralization where suppose N banks enter simultaneously, locate uniformly over the unit circle $[0, 1]$, and set deposit rates r_D^n , $n = 1, \dots, N$ to maximize profits given monopolistic competition. Assume the loan interest rate r^L is exogenously given (say due to competition in the loan market from the rest of the world).

To determine the supply of deposits to bank n (i.e. the set of households (hhs) in $[0, 1]$ who want to deposit at bank n), consider a hh who is evaluating whether to deposit in bank n or bank $n + 1$. The hh is distance $\hat{x}^n$ from bank n and distance $(\frac{1}{N} - \hat{x}^n)$ from bank $n + 1$. The depositor who is distance $\hat{x}^n$ from bank n is indifferent between going to bank n or bank $n + 1$ if the return she gets net of distance costs from the two different banks are equal or:

$$r_D^n - \gamma \hat{x}^n = r_D^{n+1} - \gamma \left(\frac{1}{N} - \hat{x}^n \right) \iff \hat{x}^n = \frac{1}{2N} + \frac{r_D^n - r_D^{n+1}}{2\gamma}.$$

¹⁸Again, an excellent exposition is in Freixas and Rochet (2008). Nielsen and Weinrich (2023) provide a Salop model to study the impact of capital requirements on market structure.

¹⁹Total travel costs is simply the area of 2 triangles (with base $1/2N$ and height $\gamma/2N$) around each of the N banks or $\frac{1}{2} \cdot \left(\frac{1}{2N}\right) \cdot \left(\frac{\gamma}{2N}\right) \cdot 2N = \frac{\gamma}{4 \cdot N}$.

Since bank n draws depositors whose costs are less than going to banks $n + 1$ or $n - 1$, bank n 's potential supply of deposits is the sum of those 2 distances:

$$D_n^s(r_D^n, r_D^{n+1}, r_D^{n-1}) = \frac{1}{N} + \frac{2r_D^n - r_D^{n+1} - r_D^{n-1}}{2\gamma}.$$

Note that the supply of deposits to bank n rises if it offers a higher interest rate r_D^n but falls if bank n 's distant competitors raise their interest rates (drawing customers away from bank n). This latter effect is where “competition” enters into monopolistic competition.

Profits of bank n are

$$\Pi_n = \max_{r_D^n} (r - r_D^n) \cdot D_n^s(r_D^n, r_D^{n+1}, r_D^{n-1}) \quad (16)$$

Given that banks move simultaneously in setting deposit interest rates, the system of first order conditions imply:

$$D_n^s(r_D^n, r_D^{n+1}, r_D^{n-1}) = (r - r_D^n) \cdot \frac{dD_n^s}{dr_D^n} \iff (r - r_D^n) = \frac{\gamma}{N} + \frac{2r_D^n - r_D^{n+1} - r_D^{n-1}}{2} \quad (17)$$

Since we assumed that all banks are identical in their technologies and all depositors are spread equally across locations, (17) has to hold for all $n = 1, 2, \dots, N$. Hence the symmetric solution is given by

$$r_D^n = r_D^{n+1} = r_D^{n-1} = r - \frac{\gamma}{N} \iff r - r_D^n = \frac{\gamma}{N}. \quad (18)$$

Equation (18) implies an interest margin that depends on how close its competitors are. Hence, (18) into (16) implies profits in a symmetric equilibrium are given by

$$\Pi_n^* = \left[r - \left(r - \frac{\gamma}{N} \right) \right] \cdot \left[\frac{1}{N} \right] = \frac{\gamma}{N^2} \quad (19)$$

Thus, monopolistically competitive banks earn positive profits in equilibrium, decreasing in the number N of banks in the industry.

Given it costs F to set up a bank, from (19) a bank will only enter if profits net of set-up costs weakly exceed 0 or $\frac{\gamma}{N^2} - F \geq 0$, with equality by the free entry condition. In that case

$$N^{MC} = \sqrt{\frac{\gamma}{F}} \quad (20)$$

This condition implies that if F is arbitrarily small, we approach perfect competition).

Comparing the socially optimal number of banks in (15) with the decentralized number of banks (20), we see

$$N^* = \frac{1}{2} \sqrt{\frac{\gamma}{F}} < \sqrt{\frac{\gamma}{F}} = N^{MC}$$

so monopolistic competition generates “too many” banks from a social perspective. Each bank in the free market chases profits and doesn't take into account the negative externality it imposes on other banks (stealing its competitors' depositors) .

In summary, this is a very important and powerful insight from theory that can help us understand tradeoffs in the data that require policy intervention. There may be a negative externality generating excessive competition which in combination with the last section, has implications for financial stability. The tradeoff from more concentration is the potential for market inefficiency owing to market power.

5 A Simple Dynamic Framework

Here we will retain the assumption of symmetry as in the static models of the last section in order to maintain tractability in a more general framework that adds dynamics, endogenous market structure, and agency frictions to address risk taking in a banking industry equilibrium model in order to understand possible trade-offs between competition and stability studied in Corbae and Levine (2024). Similar to the beautiful 3 equation New Keynesian model of Clarida, Gali, and Gertler (1999), the simple framework presented here is completely characterized by 3 equations in 3 endogenous unknowns: the degree of bank loan portfolio risk taking, the Cournot choice of deposit inflows to fund its loans, and industry market structure consistent with free entry. Given here I focus on a symmetric Cournot equilibrium, I defer the important role of heterogeneity to Sections 6-8.

There are three key frictions in our model: (i) Moral hazard induced by deposit insurance and limited liability so that risk based pricing to incentivize good behavior is absent; (ii) Imperfectly competitive granular banks whose strategic financing decisions spill over to other banks and hence can affect stability; and (iii) Agency conflicts between bank managers and shareholders to assess regulation aimed at mitigating corporate governance frictions described in Laeven and Levine (2009).

5.1 Environment

As in Allen and Gale (2004), there is a risky technology indexed by $S \in [0, 1]$. Since we calibrate the parsimonious model to a small set of data moments, we parameterize the technologies. For each unit input, the technology yields $A \cdot S$ with probability $p(S)$ and yields 0 otherwise. The parameter A can be thought of as a demand or productivity shifter. We assume $p(S) = 1 - S^\eta$, where $\eta \geq 1$. We think of η as parameterizing the monitoring technology; a higher η implies better monitoring resulting in less risk. If $Z \geq 0$ units are invested in the technology, then expected output $Y \equiv p(S) \cdot S \cdot A \cdot Z$. The cost of investing Z units is given by the strictly convex function $\tilde{b} \cdot Z + \tilde{\gamma} \cdot Z^2$, which assures an interior solution.

In the decentralized version of this economy, there are N banks that Cournot compete for insured deposits. After an initial equity injection, E_i , to finance the fixed entry costs κ of starting bank i , loans are financed by deposits since we assume seasoned equity issuance is sufficiently costly (i.e., for bank i , $L_i = D_i$). The total supply of deposits is given by $Z = \sum_{i=1}^N D_i$ with inverse deposit supply function given by $r_D(Z) = b + \gamma Z$. Thus when a given granular bank Cournot competes with other banks for deposits, it recognizes its impact on its own cost of external funding, but does not internalize the negative impact on other banks. We think of b as parameterizing the liquidity services of deposits (i.e. if

negative, households supply funds to banks at possibly dominated interest rates to enjoy their convenience yield) as in Krishnamurthy and Vissing-Jorgensen (2012) and Begenau (2020). We think of γ as parameterizing the costs of attracting external funds competing with other banks and non-banks (e.g. more competition from non-banks raise the marginal cost γ of attracting funding).

A bank manager chooses the riskiness of the loan portfolio S_i and its scale D_i to maximize the discounted profits of the bank subject to a leverage constraint that $\frac{D_i}{E_i} \leq \lambda$. Limited liability implies that if a bank is insolvent, it does not pay its depositors. As in Allen and Gale (2004), we assume the bank portfolio is well diversified and hence the state of failure corresponds to an aggregate financial crisis. We assume that there is an additional cost α to obtaining external funds that is controlled by monetary authorities (which may be interpreted as a Fed Funds rate). The manager discounts cash flows at rate β . Shareholders with linear preferences and discount factor δ make an initial equity injection to cover the entry cost (i.e., $E_i = \kappa$). The possibility of agency conflicts between the manager and equity holders is captured by $\delta \geq \beta$. We assume a large number of managers, so they take compensation as given. Managers receive a constant fraction f of the earnings of the bank while equity holders receive a fraction $1 - f$. Static preferences of the manager are given by $u(c_M) = \psi_M c_M$ while preferences of equity holders are given by $u(c_E) = \psi_E c_E$. To keep notation simple, we let $\psi_M = f^{-1}$ and $\psi_E = (1 - f)^{-1}$.

5.2 Equilibrium

We solve for a Cournot equilibrium in a decentralized imperfectly competitive banking industry with limited liability and agency frictions. Given such frictions, there is a role for policy to mitigate these frictions and bring the decentralized allocation closer to the “socially efficient” levels of risk and investment chosen by a social planner.

Assuming limited liability, incumbent bank i ’s static profit function is given by

$$\pi_i(S_i, D_i; N) = p(S_i) \left[A \cdot S_i - \left(r_D \left(\sum_{i=1}^N D_i \right) + \alpha \right) \right] D_i. \quad (21)$$

Since an incumbent manager maximizes the present value of the solvent bank at discount rate β , the dynamic problem of bank i is given by

$$V_i(N) = \max_{S_i, D_i} \pi_i(S_i, D_i; N) + \beta p(S_i) V_i(N') \quad (22)$$

subject to

$$\frac{D_i}{E_i} \leq \lambda \quad (23)$$

where N' denotes the number of banks next period.

Recognizing a given manager solves problem (22)-(23) to generate a sequence of cash flows $\pi_i(S_i, D_i; N)$ each period, shareholders with discount rate δ will inject equity to fund bank i entry provided

$$E_i(N) \equiv \frac{\pi_i(S_i, D_i; N)}{1 - \delta p(S_i)} \geq \kappa. \quad (24)$$

Free entry requires (24) holds with equality.

In what follows we focus on symmetric Markov Perfect Cournot equilibria where $S_i = S^*$ and $D_i = D^*$ for all i . In that case, (22) and (24) with equality implies

$$V(N) = \frac{[1 - \delta p(S^*)]}{[1 - \beta p(S^*)]} \cdot E(N), \quad (25)$$

so that there is an “agency wedge”

$$w(S^*) \equiv \frac{[1 - \delta p(S^*)]}{[1 - \beta p(S^*)]} \quad (26)$$

between managerial value of the firm and shareholder value. In particular, when managers are myopic relative to shareholders (i.e $\beta < \delta$), the wedge $w(S^*) < 1$ and shareholders value the firm more than the manager (i.e., $V(N) < E(N)$).

These agency conflicts have important implications for how leverage affects risk taking. In particular, the two first order conditions with respect to (S_i, D_i) in a symmetric equilibrium where the leverage requirement is non-binding can be written

$$p(S^*) = -\frac{p'(S^*)}{A} \cdot \left[R^* + \beta \cdot \frac{E(N)}{D^*} \cdot w(S^*) \right], \quad (27)$$

$$R^* = \frac{r'_D(Z)}{N} \cdot Z, \quad (28)$$

where $R^* \equiv (A \cdot S^* - (r_D(Z) + \alpha))$ denotes the interest margin and $Z \equiv N \cdot D$ denotes aggregate deposits. Since $-p'(S^*) > 0$, (27) implies that ceteris paribus the probability of success is inversely related to leverage and agency conflicts. Importantly, (27) shows there is an interaction between leverage and agency. Finally, (27) implies that, ceteris paribus, constraints on the amount of leverage the bank can take on (i.e., leverage requirements) will raise the likelihood of success. Finally, equation (28) says that, for a given amount of aggregate deposits Z , the interest margin R^* is declining in competition $\frac{r'_D(Z)}{N} = \frac{\gamma}{N}$.

We focus on symmetric steady state Cournot equilibria. What is important about the framework is its simplicity; one only needs to solve for three simultaneous equations in three unknowns. Specifically,

- i. The first order condition (27) for risk taking S key to pinning down the probability of success.
- ii. The first order condition (28) for deposit funding D key to pinning down deposits.
- iii. The free entry condition ((24) with equality) key to pinning down market structure N .

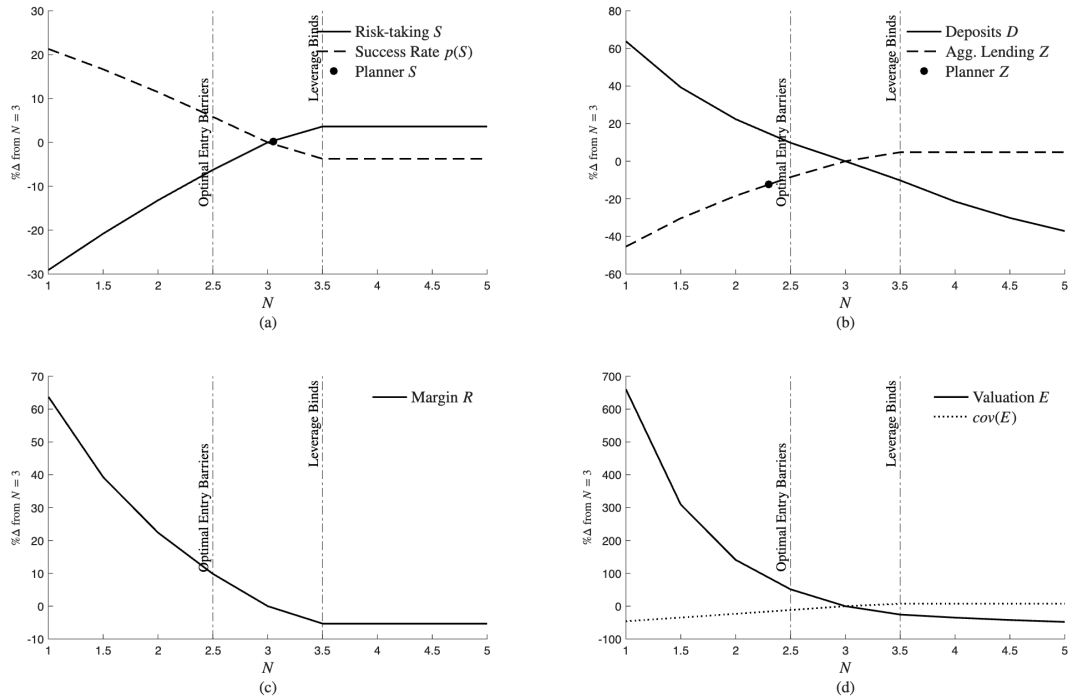
5.3 Some Key Insights

After calibrating our model to data, we use the model to address a battery of questions about the impact of regulatory and monetary policy. Specifically, the calibration uses the bank-state-time dependent data constructed by Jiang, Levine, and Lin (2023) where the average number of banks using their definition of a market is $N = 3$, which we take to be the benchmark level of concentration when calibrating our model.

5.3.1 On Competition, Efficiency, and Stability

In the benchmark model, individual banks make strategic decisions in a decentralized environment with limited liability and agency conflicts (as opposed to a social planner selecting optimal levels of risk and lending in a frictionless environment using only the technology described in the first paragraph of Subsection 5.1. Figure 2 from Corbae and Levine (2024) depicts percentage deviations of risk taking (S) and aggregate lending (Z) from the benchmark vis-a-vis levels: (i) chosen by the social planner, (ii) that arise in a less competitive economy (where $N = 1$), and (iii) that arise in a more competitive economy (where $N = 5$). Furthermore, it shows the percentage deviations of risk taking and aggregate lending that arise when a regulator optimally chooses entry barriers (κ) to minimize equally-weighted deviations of bank risk taking and output from the social planner's efficient levels.²⁰

Figure 2: Competition, Stability, and Efficiency



Source: Corbae and Levine (2024). **Note:** Percent deviations from the benchmark ($N = 3$): Less Competitive ($N = 1$), More Competitive ($N = 5$), Optimal Policy ($N = 2.5$).

What are the predictions from the changes in market structure in Figure 2. In these experiments, we choose the level of entry costs κ consistent with a given market structure. For example, κ is lower for the benchmark with $N = 3$ than the κ consistent with the monopoly case where $N = 1$. We find that when we choose an even lower κ consistent with

²⁰Given space constraints, see Corbae and Levine (2024) for the statement of the planner's problem and discussion of the regulator's problem, as well as deposit funding implications of instability (i.e. the government's budget constraint).

$N = 5$, the equity value of banks in this more competitive environment is sufficiently low that the leverage constraint $D/E \leq \lambda$ binds, illustrated in Figure 2. The fact that more competitive environments are likely to result in binding regulatory constraints is consistent with the modeling environment in Begenau (2020).

Some key insights from the counterfactuals are:

1. There is a positive monotonic relation between competition N and risk taking scale S until the leverage constraint binds, constraining risk taking in panel a. This monotonic relation translates into lower probabilities of success in more competitive economies (i.e. more competition, more financial instability). There is an important interaction between different branches of regulation: leverage constraints can help mitigate risks of more competition.
2. While the social planner takes less risk than the benchmark, despite the agency problems and limited liability, banks in the less competitive $N = 1$ economy actually take less risk than what the social planner would choose. That is, too little competition may generate inefficiently low risk taking as a monopolist bank attempts to guard its charter value. Since the choices of the decentralized bank differ from the social planner's choice, there is a role for a policymaker to intervene.
3. Relative to the social planner, there is “over-investment” (Z) in the benchmark and more competitive economies and “under-investment” by a monopolist in panel b. Investment depends not only on the number of banks but also the “size” (D) of each bank. Banks are monotonically smaller the higher the degree of competition.
4. Interest margins are monotonically decreasing in the level of competition (i.e. more competition, more market efficiency in panel c).
5. Leverage is monotonically increasing in the degree of competition until the regulatory constraint binds since equity values E in panel d fall faster than deposits D in panel b.
6. The economy is more volatile in competitive environments. The coefficient of variation of both output and equity value are increasing in the degree of competition in panel d.

Given that there is excessive risk taking and over-investment in the benchmark, there is room for a regulator to adjust entry barriers to help alleviate this inefficiency. As evident from the previous findings, the level of risk taking and aggregate investment undertaken by the social planner lies between that of the monopoly ($N = 1$) and the benchmark ($N = 3$) market structures. To analyze how anti-trust policy makers would choose the optimal entry barriers κ , we need to define “optimal”. Here, we have the policy maker choose the level of entry barriers κ that minimizes equal weighted deviations from the levels of risk taking and output chosen by the social planner. It is clear from Figure 2 that by optimally choosing a higher entry barrier κ induces the “number” of banks to fall to 2.5, implementing a market structure closer to that of a duopoly.

5.3.2 Governance and Regulatory Policy Counterfactuals

Given the interaction between agency frictions embodied in the wedge w defined in (26) and leverage D/E in the optimal choice of risk taking in (27), the model makes the prediction that there can be gains to using multiple regulatory channels (governance and leverage restrictions) to mitigate risk taking. In particular, we can use our model as a laboratory to analyze the short run (where N is fixed at the benchmark level) and long run (where N changes to reflect the impact of a given policy on bank profitability and entry) from policies which decrease the agency wedge (setting manager discount factor β closer to shareholder discount factor δ) as well as tightening the leverage constraint (lowering λ in (23)).²¹ See Figure 3 in the Online Appendix to view short and long-run equilibrium outcomes for the following counterfactual experiments.

Beginning with evaluating policies that mitigate the agency wedge, Corbae and Levine (2024) show that such policies encourage less risk taking and less leverage resulting in lower aggregate lending in the short run. The ensuing increase in success probability leads to an increase in expected future profits and equity values as well as a drop in the volatility of bank equity.²² Given that the governance change leads to higher equity values, there will be entry in the long run inducing more competition in the banking industry. Despite the ensuing long run increase in competition, the banking industry is more stable in the long run (i.e. the volatility of equity drops).

Next, we analyze the impact of tightening leverage requirements (to a level which is binding relative to the unconstrained benchmark). In the short run, tighter leverage requirements lead to less risk taking and less lending/investment relative to the benchmark. With reduced leverage, external funding costs drop and interest margins rise leading to a short run rise in profits and equity values. Tighter leverage constraints achieve the goal of short run financial stability (i.e. lower volatility of bank equity). Given that the regulatory change leads to a rise in equity values in the short run, there will be entry in the long run inducing more competition. Thus tightening leverage constraints has the perhaps unintended long run consequence of creating a more competitive banking system.

Finally, we assess the possible interaction of better governance and tighter leverage regulation suggested by the manager optimality condition (27). Specifically, we run an experiment where we both reduce λ and increase β to see if there is a complementarity between the two policies (i.e. magnification of the reduction in risk taking). Owing to the highly non-linear elasticity of our agency wedge $w(S_n^C)$ with respect to risk-taking S_n^C in (27), leverage and managerial myopia do not generate cross-partials of the same sign everywhere in the parameter space. Thus, it becomes a quantitative question. Under our benchmark calibration, the model predicts that for sufficiently tight leverage constraints, such a complementarity exists.

²¹Corbae and Levine (2024) implement short and long run equilibria following a policy change in a probabilistic manner. Specifically, given an unanticipated policy change, in the short run decision makers react optimally to the policy change anticipating the new long run market structure N' consistent with the policy change occurs with probability ζ . The probability can be chosen to match the duration of the transition between the two steady states associated with the different policies.

²²The decrease in lending outweighs the increase in success probability to generate lower expected output but also lower variation of output.

In the next subsection we test for this predicted interaction effect in the data.

5.3.3 Validation of Model Predictions

We can use our simple model to make predictions about how policies affect risk taking, charter values, and equity volatility that we can test in the data. Using the dataset constructed in Jiang, Levine, and Lin (2023) that our model parameters were calibrated to match, in this subsection we present a subset of regression results found in Section 4 of Corbae and Levine (2024) in order to validate our model predictions on how better governance and tighter leverage requirements affect equity volatility consistent with a risk taking channel.²³

Specifically, we assess the implications of competition and governance on bank risk taking using the following regression specification:

$$Y_{bst} = \gamma_C \cdot C_{bst} + \gamma_L L_{bst-1} + \gamma_A A_{bst-1} + \gamma_G G_{bst} + \gamma_I G_{bst} \cdot L_{bst-1} + \theta_b + \theta_{st} + \varepsilon_{bst} \quad (29)$$

For BHC b , headquartered in state s , in year t , Y_{bst} is *Bank Risk*, which equals the natural logarithm of the standard deviation of daily stock returns. *Bank Competition* (C_{bst}) is a measure of regulatory-induced competitive pressures facing BHC b in state s , in year t that is constructed using a gravity model associated with branching across state lines around the Riegle-Neal Act.²⁴ In addition, we include several time-varying BHC-level controls, measured in period $t-1$. Specifically, *Leverage – Lagged* (L_{bst-1}) equals the BHC’s debt to equity ratio one-year lagged and *Ln(Total Assets)–Lagged* (A_{bst-1}) equals the natural logarithm of the BHC’s total assets one-year lagged. To address governance frictions, we include % *Institutional Ownership* (G_{bst}) which equals the percentage of shares held by institutional investors (capturing the idea that institutional investors are more informed than individual investors and better able to incentivize executives to focus more on the long-run value of the bank). Finally, the regressions control for bank (θ_b) and state-year (θ_{st}) fixed effects, and ε_{bst} is the error term.

Table 12 in the Online Appendix presents the empirical results along with the model predictions from section 5.3.2. We find that an intensification of competition increases bank risk consistent with predictions of the model. Furthermore, the estimated economic impact of competition on BHC risk is large. For example, consider a BHC that experiences a change in *Competition* from the 25th percentile (1.14) to the 75th percentile (1.96) of the sample distribution, an increase of 0.82. Our estimates suggest that Bank Risk would rise by 0.49 and be 15% greater in the more highly competitive environment (3.72) than in the less competitive environment (3.23). With respect to the other control variables, the results confirm the predictions of our model. Consistent with the views that larger banks are better diversified (Goetz, Laeven, and Levine Goetz et al. (2016)) and perhaps also too-big-to-fail,

²³We also confirm the interaction of bank competition and contractionary monetary policy in Table 4 of Corbae and Levine (2024).

²⁴For the case of banks, the gravity model assumes that the costs to a bank of establishing and effectively managing an affiliate increase with the geographic distance between the BHC’s headquarters and the affiliate. Consistent with this gravity view of bank behavior, Goetz et al. (2013) and Goetz et al. (2016) show that BHCs are more likely to expand into geographically close markets. The gravity model has important implications for the competitive pressures triggered by interstate bank deregulation.

we find that bank size, $\ln(\text{Total Assets})\text{-Lagged}$, is inversely related to risk. Consistent with the view that more levered banks are more fragile, we find that Leverage-Lagged is positively associated with risk.

Importantly, our model offers insights into how leverage requirements and regulations on executive incentives interact to shape excessive risk taking by banks consistent with the regression results in Table 12. In particular, the model explains how (under plausible parameterizations) a tightening of leverage requirements will have a bigger risk-reducing effect when bank executives are more concerned about the long-run profitability of the bank and hence less myopic; forcing banks to be equity financed reduces excessive risk taking more if bank executives are more concerned about the equity value of the bank. The model also indicates that regulations that induce bank executives to focus less on short-run bonuses and more on the longer-run charter value of the bank will have a larger risk-reducing effect when the bank is less levered. The policy implication is potentially first-order; the result stresses that leverage requirements and regulations on executive incentives are reinforcing.

5.4 Further Results

In Corbae and Levine (2024) we also study how competition and stability interacts with monetary policy (see a summary in Section A.3.1 in the Online Appendix). Section 3.3 of that paper shows how the simple model can be extended to consider the implications for competition and stability from: (i) Too-Big-To-Fail Policy Experiments, (ii) the rise of Shadow Banks and Regulatory Arbitrage, (iii) Fintech, (iv) Productivity Shocks, (v) Contagion and Bank Runs.

Quantitative Heterogeneous Banking Models

To keep the models simple, Sections 4 and 5 focused on symmetric equilibria. The data described in Section 3 documents that there can be large differences in the size distribution of banks and that this can have important implications for financial stability. For instance, there is support for granularity in the banking sector, where an idiosyncratic shock to funding one of the largest 35 banks in the U.S. can have a sizeable impact on aggregate lending in an industry with over 4000 banks. The next sections 6-8 introduce idiosyncratic shocks to models with granular banks consistent with the data in Section 3 that can be used as a laboratory to assess the impact of regulatory policies discussed in Section 2 on competition, stability, and efficiency.

6 On the Bank Deposit Channel

Here we begin with Egan, Hortaçsu, and Matvos (2017) focusing on the liability side of the bank balance sheet. Their paper is a quantitative heterogeneous bank version of Diamond and Dybvig (1983).²⁵ It addresses several important questions about existence of run equi-

²⁵See also Gertler and Kiyotaki (2015) for a dynamic quantitative macroeconomic model of bank runs with a representative bank since their environment admits a leverage multiple which is independent of bank specific factors and they consider the case where the incentive constraint for banks is binding.

libria and the quantitative significance of spillovers between granular (the top 16) banks in a model with parameters estimated to match U.S. deposit data. Given deposit insurance, it is not the insured depositors who run, but instead uninsured depositors are sensitive to bank default which induce runs.

6.1 Bank Run Environment

The essential elements of their model are as follows. Time is discrete. There are M^I consumers for insured deposits, M^N consumers for uninsured deposits, and K banks.

The timing in any period is as follows:

1. Banks set interest rates for insured and uninsured deposits, $i_{k,t}^I$ and $i_{k,t}^N$.
2. Depositors of each type choose which of the K banks to deposit with.
3. Banks invest deposits after which shocks to their loan portfolio are realized.
4. Banks choose whether to repay deposits and the coupon on the long term debt they owe or default (i.e. whether there is a seasoned equity injection to maintain charter value).

Consumer preferences are modeled as in the industrial organization, discrete choice (logit) literature. Risk neutral consumer j 's utility from funding bank k at time t via uninsured deposits is given by

$$u_{j,k,t}^N = \alpha^N i_{k,t}^N - \rho_{k,t} \gamma + \delta_k^N + \epsilon_{j,k,t}^N.$$

where $i_{k,t}^N$ is the interest rate on uninsured deposits, γ represents the loss in utility when bank k defaults with probability $\rho_{k,t}$, δ_k^N is a bank specific fixed effect, and $\epsilon_{j,k,t}^N$ is an i.i.d utility shock drawn from a Type 1 extreme value distribution. The parameter γ will be important for funding decisions leading to crises. Insured depositors have similar preferences but they do not lose utility from default (i.e. there is deposit insurance):

$$u_{j,k,t}^I = \alpha^I i_{k,t}^I + \delta_k^I + \epsilon_{j,k,t}^I.$$

Banks maximize their equity value by competing on deposits. The lending side of the bank balance sheet is exogenously given (something that will be relaxed in Sections 7 and 8); that is, given its deposit funding, bank k receives the return on its loan portfolio net of non-interest costs, denoted by $R_{k,t} \sim N(\mu_k, \sigma_k)$, which is i.i.d. over time. There is an additional cost c_k to fund insured deposits. It is assumed that to fund entry (which is exogenous since it occurred in the past), banks issued a consol bond in the past, so they need to repay b_k every period.

The profit of bank k at time t is given by

$$\pi_{k,t} = M^I s_{k,t}^I (R_{k,t} - c_k - i_{k,t}^I) + M^N s_{k,t}^N (R_{k,t} - i_{k,t}^N).$$

where $s_{k,t}^i$ is the market share of bank k at time t in deposit market $i = I, N$. At time t , the bank uses its net cash inflow $\pi_{k,t} - b_k$ to pay dividends; by assumption the bank cannot retain earnings. If $\pi_{k,t} - b_k < 0$, equity holders can choose to finance the loss or not. In the case of default, equity holders loose the claim on the future dividends, the bank is sold, and the proceeds are used to repay depositors and bondholders. It is assumed the failed bank is replaced by a “replica” bank.

6.2 Bank Run Equilibrium

Facing interest rates $i_{k,t}^I$ and $i_{k,t}^N$, each type of consumer chooses which bank to deposit their funds with in order to maximize utility. Consistent with the discrete choice literature, the insured and uninsured market shares of bank k are given by the standard logit form:

$$s_{k,t}^I(i_{k,t}^I, \mathbf{i}_{-k,t}^I) = \frac{\exp(\alpha^I i_{k,t}^I + \delta_k^I)}{\sum_{l=1}^K \exp(\alpha^I i_{l,t}^I + \delta_l^I)}, \quad (30)$$

$$s_{k,t}^N(i_{k,t}^N, \mathbf{i}_{-k,t}^N, \rho_{k,t}, \boldsymbol{\rho}_{-k,t}) = \frac{\exp(\alpha^N i_{k,t}^N - \rho_{k,t}\gamma + \delta_k^N)}{\sum_{l=1}^K \exp(\alpha^N i_{l,t}^N - \rho_{l,t}\gamma + \delta_l^N)} \quad (31)$$

It is evident from (30) and (31) that bank k 's market share is increasing in its deposit rate and decreasing in its likelihood of failure.

A bank defaults if its current profits are sufficiently negative relative to the charter (equity continuation) value of the bank:

$$\pi_{k,t} - b_k + \frac{1}{1+r} E_k < 0. \quad (32)$$

The optimal default policy satisfies a threshold property; default if $R_k < \bar{R}_k$ where $\bar{R}_k$ satisfies (32) with equality.

The Bellman equation for bank k is

$$E_k = \max_{i_k^I, i_k^N} \int_{\bar{R}_k}^{\infty} \left[M^I s_k^I(i_k^I, \mathbf{i}_{-k}^I) (R_k - c_k - i_k^I) + M^N s_k^N(i_k^N, \mathbf{i}_{-k}^N, \rho_k, \boldsymbol{\rho}_{-k}) (R_k - i_k^N) - b_k + \frac{1}{1+r} E_k \right] dF(R_k) \quad (33)$$

After computing the conditional expectation in (33) analytically, the two first order conditions are given by:

$$i_k^I : \mu_k + \sigma_k \lambda \left(\frac{\bar{R}_k - \mu_k}{\sigma_k} \right) - c_k - i_k^I = \frac{1}{\alpha^I (1 - s_k^I(i_k^I, \mathbf{i}_{-k}^I))}, \quad (34)$$

$$i_k^N : \mu_k + \sigma_k \lambda \left(\frac{\bar{R}_k - \mu_k}{\sigma_k} \right) - i_k^N = \frac{1}{\alpha^N (1 - s_k^N(i_k^N, \mathbf{i}_{-k}^N, \rho_k, \boldsymbol{\rho}_{-k}))}. \quad (35)$$

Equations (34) and (35) relate bank markups positively to market shares. They make clear that that determinants of the loan return and market shares affect the interest rate that

banks pay on their deposits. Further, since the market share on uninsured deposits depends on the probability of default, interest rates on those funds depend on beliefs about the solvency of each bank.

An equilibrium of this model consists of: (i) default probabilities, (ii) default thresholds, (iii) interest rates on insured and uninsured deposits, and (iv) market shares such that:

1. Consumers choose where to deposit optimally given by (30) and (31),
2. Banks choose default threshold optimally:

$$\begin{aligned}
& -M^I s_{k,t}^I (\bar{R}_k - c_k - i_{k,t}^I) - M^N s_{k,t}^N (\bar{R}_k - i_{k,t}^N) + b_k = \\
& (M^I s_{k,t}^I + M^N s_{k,t}^N) \left[\mu_k - \bar{R}_k + \sigma_k \lambda \left(\frac{\bar{R}_k - \mu_k}{\sigma_k} \right) \right] [1 - F(\bar{R}_k)]
\end{aligned} \tag{36}$$

3. Banks choose interest rates optimally given in (34) and (35)
4. Rational expectations: consumers' beliefs about the probability of bank default are correct in equilibrium:

$$\rho_k = P(R_k \leq \bar{R}_k) = \Phi \left(\frac{\bar{R}_k - \mu_k}{\sigma_k} \right) \tag{37}$$

where $\lambda(x) = \frac{\phi(x)}{1-\Phi(x)}$ is the inverse Mills ratio, and $\phi(x)$ and $\Phi(x)$ are the probability density and cumulative distribution functions of the Normal distribution, respectively. Thus, an equilibrium is given by $6K$ equations in $6K$ unknowns. It is possible to reduce the number of variables to $3K$ by substituting (30), (31), and (37) into (34), (35), and (36).

As in Diamond and Dybvig (1983), there can be multiple equilibria in the Egan, Hortaçsu, and Matvos (2017) model depending on parameter values, especially the aversion to losses from bank failure γ . The fixed point behind the bad equilibrium is given by:

1. Uninsured depositors suddenly believe (a sunspot) that a bank may default.
2. They reduce their demand for deposits from that bank.
3. Lower demand decreases the profitability of the bank.
4. Lower profitability increases the probability of default, which is consistent with the belief.

If uninsured depositors do not care about default (i.e. $\gamma = 0$), there is no reason to reduce demand and the bad equilibrium fails to exist.

This raises a fundamental quantitative question: how large is γ in the data? To answer this question, the authors use bank specific data on Credit Default Swap (CDS) spreads across time to back out ρ_{kt} and then use an instrumental variable approach to regress deposit market share of bank k on its probability of default and other controls yielding an estimate of γ .

Egan, Hortaçsu, and Matvos (2017) illustrate a fundamental source of contagion. Here, unlike the literature on contagion due to linked interbank balance sheets where a failure by a borrower bank spills over to the asset side of another bank’s balance sheet as in the bank network literature (see especially Allen and Gale (2000) and Acemoglu, Ozdaglar, and Tahbaz-Salehi (2015)), the spillover arises due to strategic interactions among big banks in the deposit market (as we will see the models of Sections 7 and 8 illustrate this channel in the loan market). Specifically, consider a bad equilibrium in which bank k suffers from a sunspot shock where depositors believe that uninsured deposits will fail. In that equilibrium, the expected return conditional on not failing is higher for bank k , so it chooses to offer higher deposit rates. Due to imperfect competition, other banks respond by offering higher interest rates to try to maintain their funding. Due to higher interest rates, firms are less profitable, which spills over to higher aggregate default rates.

Table 4 in Egan, Hortaçsu, and Matvos (2017) illustrates the multiplicity. In line with this chapter’s focus on granular banks, the authors present results for five of the largest banks (Chase, Bank of America, Wells Fargo, Citibank, and Wachovia). They examine, for instance, what would have happened if Wachovia’s probability of default rose from the observed 3.3% to 52% because of a change in beliefs. In that case, uninsured interest rates at Wachovia rise from 2.2% to 14% and even insured rates rise from 2% to nearly 9% as they try to attract more lower cost insured deposits. To compete for deposits, the other big banks must raise their interest rates, cutting profit margins which induces a higher probability of default for all banks.

7 Bank Mergers

Here we focus on the role that Riegle-Neal played in the changing bank market structure evident in Figure 1 between two periods with relatively stable levels of concentration albeit at different levels; a low level from 1983-1994 and a high level from 2010-2023.²⁶ Consistent with the data we discussed in Section 3 that 96% of bank mergers involve an acquirer that is significantly larger than its target and the median acquirer has over seven times more assets than the target, Corbae, D’Erasmus, and Smith (2025) build on the non-financial dominant-fringe model of Gowrisankaran and Holmes (2004) to assess the financial stability and welfare consequences of the bank merger wave. To be consistent with the banking data in Section 3, we add the following elements to their model: fixed operating costs and differences across bank type of extending loans inducing increasing returns unlike their constant returns to scale production function, a non-bank sector competing with our banking sector to address issues of regulatory arbitrage and possible changes in technology as in Buchak et al. (2018), and entry and exit to address issues of financial stability and how mergers affect the size of the banking industry.

We calibrate the model to match the aggregate dynamics of market shares in Figure

²⁶Aguirregabiria, Clark, and Wang (2016) argue that because of economies of density and merger costs, smaller banks did not take advantage of the benefits of geographic deposit risk diversification and that larger banks, with relatively smaller merging costs, expanded out of their home state but not to reduce their geographic risk, but to increase expected log-deposits per branch.

1. Once calibrated and validated, we have a laboratory to conduct counterfactuals such as “What would have happened if Riegle-Neal was never enacted?” With changes across the financial industry post-crisis like Dodd-Frank regulations and the rise of competition from the nonbank sector (which itself may be a consequence of regulatory arbitrage), would there have been more or less bank failures? Have mergers contributed to allocative inefficiency? How has increased concentration affected the bank lending channel and the effectiveness of monetary policy? Further, we can use the laboratory to study optimal dynamic bank merger regulation.²⁷ Nocke and Whinston (2022) document screening thresholds (across HHI level and change) in the 1982 and 2010 Horizontal Merger that we use to guide regulatory costs associated with mergers.

7.1 Merger Environment

Corbae, D’Erasmus, and Smith (2025) consider an infinite horizon, discrete time model but to save on notation we use the nomenclature that variable x_t is denoted x and x_{t+1} is denoted x' . Each period begins with a measure γ of identical atomless (i.e. measure zero) fringe banks each with D_f deposits as well as a granular (representative) dominant bank with D_d deposits. While we take D_f as parametrically given, D_d can expand endogenously through the acquisition of fringe banks. Specifically, to expand its deposit base, the dominant bank can make a take-it-or-leave-it (TOLI) offer to a measure $(\gamma - \Gamma)$ of fringe banks. Assuming all fringe banks accept, the measure of fringe banks falls from γ to Γ and the deposits of the dominant bank increases by $(\gamma - \Gamma)D_f$. The dominant bank chooses the size of the merger Γ to maximize its equity value. We take the cost of deposits, r_D , as exogenous calibrated to the average deposit rate and instead focus on the endogenous determination of interest rates in the imperfectly competitive loan market.

Following the merger stage, the dominant bank, the fringe banks, and the representative non-bank Cournot compete in the loan market. Both types of banks face transitory, idiosyncratic shocks to their deposits, denoted ϵ_i drawn from distribution $g_i(\epsilon_i)$. Having observed their shock, banks choose between allocating their deposits to the loan market or securities, where the latter provides an exogenous risk free return of $r_A > r_D$. Banks face a fixed cost, κ_i , and the cost of loan monitoring given by $C_i(L_i)$. The presence of fixed costs can create increasing returns along the lines of Diamond (1984). Ex-ante identical entrepreneurs can demand one-period loans in order to fund a risky project. An entrepreneur first decides whether they want to fund their project given an outside option ω drawn from a uniform distribution and then makes a discrete choice over whether to borrow from a bank B or a nonbank N . Using the discrete choice approach from IO, these borrowers create an aggregate demand for loans. Dominant banks move first followed by the fringe banks and the nonbank. Market clearing and the nonbank’s first-order condition determine equilibrium loan rates (r_B^L, r_N^L) .

²⁷Optimal merger regulation has been considered for non-financial firms in Nocke and Whinston (2010), where they focus on a model of Cournot competition with constant returns to scale, and without financial stability concerns and in Mermelstein et al. (2020), where non-financial firms can reduce costs through investment or through mergers. A model where banks grow through investment has been studied in Corbae and D’Erasmus (2020) who use a “ladder” investment technology like that in Besanko and Doraszelski (2004).

Following the loan stage, aggregate loan defaults (θ') and bank specific chargeoffs (λ') are realized in the payoff stage. We assume that θ' follows a persistent Markov process with distribution $G(\theta', \theta)$. Further, we assume that $\lambda'_N = \bar{\lambda} \in [0, 1]$ and that $\lambda'_d, \lambda'_f \in [0, 1]$ is iid over time and across fringe banks distributed with a truncated normal $F_i(\lambda; \bar{\lambda}, \sigma_{\lambda,i})$ where $\bar{\lambda}$ and $\sigma_{\lambda,i}$ denote the mean and the standard deviation of the distribution, respectively. We assume banks with non-negative profits pay them as dividends (there are no retained earnings, which will be explored in Section 8 of this chapter). Banks with negative profits can choose to exit or inject costly equity, given by ψ_i , to remain in the market. New banks can choose to enter the fringe at cost K_f . Endogenous exit and entry decisions, with associated measures γ_e and γ_x , imply that the measure of fringe banks can change through time. In sum, the size of the fringe next period is given by $\gamma' = \Gamma + \gamma_e - \gamma_x$.

In summary, the timing in any period is as follows:

- M1. Merger Stage: The dominant bank makes a take-it-or-leave-it merger bid to a measure $(\gamma - \Gamma)$ of fringe banks in order to grow its deposit base.
- M2. Loan Stage: Banks observe ϵ_i shock to their deposits. Then, banks allocate their deposits to the loan market or securities. Dominant banks move first in the loan market followed by the fringe banks and nonbanks.
- M3. Payoff Stage: At the end of the period, loan defaults and charge-offs are realized. Exit and entry decisions are made and dividends paid.

7.2 Merger Equilibrium

Aggregate deposits in the banking sector are $D = D_d + \gamma D_f$. Starting in aggregate state $s = (\gamma, D_d, \theta)$, the dominant bank makes a TOLI offer to a measure $\gamma - \Gamma \geq 0$ of fringe banks to gain deposits. Assuming all fringe banks accept, the measure of the fringe banks after the merger adjusts to Γ and the deposits of the dominant bank is given by

$$D'_d = D_d + (\gamma - \Gamma)D_f \quad (38)$$

If the price of a unit of deposits in the merger stage is denoted $p(\Gamma, s)$, then the total cost of the merger deal to the dominant bank is given by $p(\Gamma, s)(\gamma - \Gamma)D_f$. Since mergers (weakly) increase the dominant bank's deposit market share, it is possible for the model to match the data fact of rising dominant bank market share and falling number of banks evident in Figure 1. The latter result - the falling number of banks - arises as the endogenous measure of fringe banks evolves in our model which is absent in Gowrisankaran and Holmes (2004).

The dominant bank chooses a deposit market share to maximize its profitability recognizing that the fringe banks must be willing to accept their offer of size $p(\Gamma, s)(\gamma - \Gamma)D_f$. Specifically, we write the dominant bank's dynamic programming problem at the beginning of the merger stage as:

$$v_d(s) = \max_{\Gamma \in [0, \gamma]} -p(\Gamma, s)(\gamma - \Gamma)D_f - H_n(\Gamma, s) + w_d(\Gamma, D'_d, \theta) \quad (39)$$

subject to

$$p(\Gamma, s)D_f \geq w_f(S) \quad (40)$$

where $H_n(\Gamma, s), n \in \{pre-RN, post-RN\}$ are regulatory costs borne by the acquirer from the merger before and after Riegle-Neal and $w_i(S)$ is the expected present discounted value of a type $i \in \{d, f\}$ bank in loan stage state $S = (\Gamma, D'_d, \theta)$. Constraint (40) incentivizes the fringe to accept the take-it-or-leave-it offer of the dominant bank.

The regulatory cost of mergers $H_n(\Gamma, s), n \in \{pre-RN, post-RN\}$ has two components. First, under pre-Riegle-Neal restrictions, banks are constrained to operate in one state. As such, the cost of merging is infinite (i.e. $H_{pre-RN}(\Gamma, s) = \infty$). There are several conditions under post-Riegle-Neal restrictions. First, banks cannot merge to greater than 10% deposit market share.²⁸ As such, the cost of merging such that the dominant bank's deposit market share exceeds 40% (since our dominant bank represents the Top 4) is infinite. Second, to capture costs associated with HHI constraints given in Nocke and Whinston (2022), we compute an implied HHI from the model and the cost associated from the merger is given by the following functional form:

$$H_{post-RN}(\Gamma, s) = \begin{cases} \infty & \text{if } \frac{D'_d}{D_d + \gamma D_f} > 0.40 \\ h(\gamma - \Gamma)^2 & \text{if } HHI > 1800 \text{ or } \{HHI < 1800 \text{ and } \Delta_{HHI} > 100\} \\ 0 & \text{otherwise} \end{cases} \quad (41)$$

where h is parametrically given measuring the intensity of merger regulation.

Let $\Gamma^*(s)$ solve (39)-(40) noting that if the dominant bank chooses not to acquire any fringe banks, then $\Gamma^*(s) = \gamma$. Since, in the loan stage, the dominant bank can use its market power to withhold loans, fringe banks recognize that the more mergers which occur, the more valuable it is to remain in the market capturing the option value of being a fringe bank. As a result, the dominant bank faces a trade-off between acquiring a few fringe banks at a low price or many fringe banks at a high price.

Following the merger stage, the dominant bank, fringe banks, and a representative non-bank compete in the loan market. Ex-ante identical entrepreneurs demand one-period loans to fund a risky project. The project requires one unit of investment from either a bank B or nonbank N . Every period, given r_B^L, r_N^L , and shock ω , entrepreneurs choose whether to invest ($\iota = 1$) or not ($\iota = 0$) to solve:

$$\max_{\{\iota\}} (1 - \iota) \cdot \omega + \iota \cdot \mathbb{E}_\epsilon[\Pi_E(\theta, r_B^L, r_N^L, \delta)] \quad (42)$$

Conditional on choosing to invest, entrepreneurs observe $\delta = \{\delta_B, \delta_N\}$ and then choose which type of lender $k \in \{B, N\}$ to borrow from to solve:

²⁸Section 101 (2) in the Riegle-Neal Interstate Banking and Branching Efficiency Act of 1994 established nationwide concentration limits and stated that a merger ...“applicant (including all insured depository institutions which are affiliates of the applicant) controls, or upon consummation of the acquisition for which such application is filed would control, more than 10 percent of the total amount of deposits of insured depository institutions in the United States.”

$$\Pi_E(\theta, r_B^L, r_N^L, \delta) = \max_{k \in \{B, N\}} \alpha \mathbb{E}_{\theta'|\theta} [\pi(r_k^L, \theta)] + \delta_k \quad (43)$$

where

$$\pi_E(r_k^L, \theta') = \begin{cases} \max\{0, R - r_k^L\} & \text{with prob } \theta' \\ \max\{0, -(\lambda' + r_k^L)\} & \text{with prob } 1 - \theta' \end{cases}.$$

The solution to (43) implies that the share of borrowers choosing a loan from a lender of type k is

$$\Psi_k(\theta, r_B^L, r_N^L) = \frac{\exp(\alpha \mathbb{E}_{\theta'|\theta} [\pi_E(r_k^L, \theta')])}{\sum_{\hat{k} \in \{B, N\}} \exp(\alpha \mathbb{E}_{\theta'|\theta} [\pi_E(r_{\hat{k}}^L, \theta')])}. \quad (44)$$

The ex-ante value for the borrower is given by:

$$U_E = \frac{1}{\alpha} \mathbb{E}_\omega[\iota] \mathbb{E}_\epsilon[\Pi_E(\theta, r_B^L, r_N^L, \delta)] \quad (45)$$

The borrower problem creates a demand system that determines for a given level of loans by the banking system L_B and nonbank interest rate r_N^L , the endogenous r_B^L that clears the loan market.

Following the loan stage, loan defaults (θ') and chargeoffs (λ) are realized. The profit of bank $i \in \{d, f\}$ is given by:

$$\pi_i(L_i, \epsilon_i; S, \theta', \lambda'_i) = [\theta' r_B^L - (1 - \theta') \lambda'_i] L_i + r^A (D_i + \epsilon_i - L_i) - r^D (D_i + \epsilon_i) - C_i(L_i) - \kappa_i. \quad (46)$$

where $C_i(L_i)$ is the cost of loan monitoring. We assume $C_i(L_i)$ takes the following form:

$$C_i(L_i) = C_{1i} L_i + C_{2i} L_i^2 \quad (47)$$

Further, we denote dividends by:

$$\mathcal{D}_i(L_i, \epsilon_i; S, \theta', \lambda'_i) = \pi_i(L_i, \epsilon_i; S, \theta', \lambda'_i) - \mathbf{1}_{\{\pi_i(\cdot) < 0\}} \psi_i(|\pi_i(L_i, \epsilon_i; S, \theta', \lambda'_i)|). \quad (48)$$

where ψ_i captures the costs of equity issuance.

The dynamic programming problem of the dominant bank can then be written as:

$$w_d(\epsilon_d, S, r_N^L) = \max_{L_d \leq D'_d + \epsilon_d} \mathbb{E}_{\theta', \lambda'_d|\theta} \left[\max_{x'_d \in \{0, 1\}} (1 - x'_d) (\mathcal{D}_d(L_d, \epsilon_d; S, \theta', \lambda'_d) + \beta v_d(s')) \right], \quad (49)$$

subject to

$$\mathcal{L}_B \equiv L_d + \Gamma \int L_f(\epsilon_f, L_d, S, r_N^L) g(d\epsilon_f) = L_B(r_B^L, r_N^L), \quad (50)$$

$$\gamma' = F(L_d, S, \theta'). \quad (51)$$

where $s' = (\gamma', D'_d, \theta')$ and the function $F(L_d, S, \theta')$ denotes the transition function of the measure of fringe banks. Equation (50) is the loan market clearing condition where the dominant bank takes into account both how their loan decision affects the fringe loan decision

and interest rates r_B^L . Equation (51) ensures that the dominant bank internalizes its impact on market structure in the future.

Given the stackelberg game in the loan market, taking the interest rates r_B^L and r_N^L as given, the problem of a fringe bank solves²⁹

$$w_f(\epsilon_f, S, L_d, r_N^L) = \max_{\ell_f \leq D_f + \epsilon_f} \mathbb{E}_{\theta', \lambda'_f | \theta} \left[\max_{\chi'_f \in \{0,1\}} (1 - \chi'_f) (\mathcal{D}_f(\ell_f, \epsilon_f; L_d, S, \theta', \lambda'_f) + \beta v_f(s')) \right]. \quad (52)$$

The solution to (52) yields $\ell_f(\epsilon_f, S, L_d, r_N^L) = L_f(\epsilon_f, S, L_d, r_N^L)$, the latter equality due to a consistency condition.

We can use our model to study banks' heterogeneous marginal propensities to lend (MPL). Heuristically, the first order condition with respect to L_i is given by:

$$\begin{aligned} \mu_i(\epsilon_i, S; \cdot) &= \mathbb{E}_{\theta', \lambda'_i \leq \lambda_i^{x'=1} | \theta} \left\{ [\theta' r_B^L - (1 - \theta') \lambda'_i] + \left[\theta' \frac{\partial r_B^L}{\partial \mathcal{L}_B} \frac{\partial \mathcal{L}_B}{\partial L_i} L_i \right] + \left[\beta \frac{\partial v_i(s')}{\partial s'} \frac{\partial s'}{\partial L_i} \right] \right\} \\ &- E_{\theta', \lambda'_i \leq \lambda_i^{x'=1} | \theta} \left\{ \left[\frac{\partial Pr(\pi_i < 0, x'_i = 0)}{\partial L_i} \frac{\partial \psi_i(|\pi_i|)}{\partial |\pi_i|} \right] + r^A + \frac{\partial C_i(L_i)}{\partial L_i} \right\} \end{aligned} \quad (53)$$

where μ_i is the multiplier on the constraint that $L_i \leq D_i + \epsilon_i$ and $\lambda_i^{x'=1}$ is bank i 's exit threshold. For fringe banks $\frac{\partial \mathcal{L}}{\partial L_i} = \frac{\partial s'}{\partial L_i} = 0$, as they do not internalize how their lending affects interest rates or the evolution of the state variables. Notice, if $\mu_i > 0$, then the bank is constrained and would lend any additional deposits. However, if $\mu_i = 0$, then the bank is unconstrained, and would not lend more if they received additional deposits. Hence, banks MPL is either 0 or 1 across different states. In certain states (e.g. $\epsilon_i < 0$), banks behave similar to "hand-to-mouth" consumers. For the dominant bank, the marginal propensity to lend also depends on how lending affects interest rates and the evolution of future states. Notice also that differences in seasoned equity issuance costs across dominant and fringe banks also influences loan supply in (53). Since lending is risky and can lead banks to negative profits, lower external finance costs for dominant banks means lower expected cost to lending (*ceteris paribus*) and hence higher loan supply.

Turning to non-bank competition, the profits of nonbank N are given by:

$$\pi_N(\ell_N, S, \theta', \lambda') = [\theta' r_N^L - (1 - \theta') \lambda' - c_N] \ell_N. \quad (54)$$

The first order condition of the non-bank with respect to ℓ_N is given by

$$r^D = E_{\theta' | \theta} [\theta' r_N^L - (1 - \theta') \lambda'] - c_N. \quad (55)$$

Banks with negative profits can choose to exit or inject equity to remain in the market. New fringe banks can enter. The entry decision satisfies:

$$v_i^e(L_d, S, r_N^L, \theta'; \gamma^e(x'_d, e'_d)) = \max_{e'_i \in \{0,1\}} (1 - e'_i) \{-K_i + v_i(s')\}. \quad (56)$$

The evolution of the mass of fringe banks then is given by

$$\gamma' = F(L_d, \epsilon_d, S, \theta') = \Gamma - \gamma^x(L_d, \epsilon_d, S, \theta') + \gamma^e(L_d, \epsilon_d, S, \theta'). \quad (57)$$

²⁹To save on notation, we neglect adding L_f and x_f as state variables in (52).

Definition 4. Taking r^A and r^D as given, a **Markov Perfect Merger Equilibrium** is a set of value functions $\{v_i, w_i\}$ and policy functions $\{\Gamma, L_i, \ell_f, x'_i, \chi'_f, e'_i\}$ for $i \in \{d, f\}$, ℓ_N , prices $\{p, r_B^L, r_N^L\}$, and transition functions for $\{\gamma', D'_d\}$ such that:

1. The pre-merger value function v_d solves (39). The merger quantity Γ maximizes (39).
2. The merger pricing function p satisfies (40).
3. The post-merger value functions w_d and w_f solve (49) and (52). The loan supply policy functions (L_d, ℓ_f) and exit decision rules (x'_d, χ'_f) maximize (49) and (52).
4. Consistency requires $\ell_f = L_f$ and $\chi'_f = x'_f$.
5. r_B^L clears the loan market (50).
6. r_N^L satisfies the shadow bank first order condition (55).
7. The mass of entrants γ^e solves the entry problem (56).
8. Transition functions D'_d and γ' are consistent with merger, entry, and exit decisions (38) and (57).

7.3 Merger Results

The model parameters are calibrated to the pre-Riegle-Neal data (1984-1993) where banks were broadly restricted to one state and the Top 4 banks, represented by our dominant bank, accounted for a relatively stable (i.e. steady state) 15% deposit market share.³⁰ We provide certain relevant statistics for the pre-Riegle-Neal long run equilibrium in column “Pre-RN” listed in Table 3 in the Online Appendix A.

To simulate Riegle-Neal, we then remove the pre-Riegle-Neal regulatory cost $H_{pre-RN}(\Gamma, s) = \infty$ and instead apply costs associated with HHI and the regulatory cap on market shares in (41) where h is calibrated to match the length of the deposit market transition post-Riegle-Neal. Figure 4 in the Online Appendix presents the transition paths for the deposit market share and numbers of fringe banks to illustrate how the deposit market evolves in our model versus the data from 1994 to 2011.

Using our model as a laboratory, we can ask what would have happened, *ceteris paribus*, to competition, market efficiency, and stability if Riegle-Neal was never implemented?³¹ Comparing the Pre-RN column with Post-RN columns in Table 3 illustrates a competition-stability tradeoff; weaker regulation created more concentration (less competition) in the banking industry (3 times greater as in the data) along with associated increases in interest

³⁰To match the relatively stable post-Dodd-Frank (2011-2019) period, the paper also recalibrates dominant and fringe bank fixed costs along with the nonbank marginal cost in order for to capture increasing regulatory costs and heightened competition from nonbanks due to financial innovation consistent with Buchak et al. (2018).

³¹Our model also allows us to examine the interaction between regulatory and monetary policy studied in Corbae, D’Erasmus, and Smith (2025).

rates (8 basis points in the short run and 16 basis points in the long run) and dominant bank markups (up to 4% higher) but lower fringe failure rates (11% lower in the short run and 7% lower in the long run). The resulting changes in Post-RN bank charter values induce increases in Tobin’s Q both for dominant acquirers and fringe targets especially in the transition.³² Further, the experiment illustrates that the merger channel leads the banking sector to shrink by 4% relative to non-banks as opposed to 27% in the data.

8 On the Bank Lending Channel

While there is a large literature in industrial organization that focuses on bank funding via deposits studied in the previous two sections, in part due to public access to extensive data from the FDIC, macroeconomists focus much attention on how monetary policy is transmitted to the real side of the economy through the “credit channel”. Bernanke and Gertler (1995) separate the credit channel into two parts associated with the balance sheets of borrowers and lenders. The “balance sheet channel” posits that the external finance premium facing a borrower depends on the borrower’s balance sheet; the stronger the balance sheet the lower the premium to borrow. A tight monetary policy affects borrower’s balance sheets in two ways: (i) directly by raising interest expenses and reducing net cash flows thereby weakening the borrower’s balance sheet and (ii) indirectly by changing collateral values of existing assets. The “bank lending channel” can be thought of as a corporate finance application of the above concept of an external finance premium arising out of the strength of bank balance sheets. Specifically, as the empirical results in Kashyap and Stein (2000) suggest, banks with little liquidity or weak balance sheets with less access to uninsured borrowing are more sensitive to tight monetary policy since it is harder for them to raise external funds than bigger banks with greater access to uninsured borrowing.³³

An important insight from this research is that both channels work through heterogeneity of borrowers and banks. Here, I focus on how bank heterogeneity works through the lending channel and point the interested balance sheet channel reader to papers such as Ottonello and Winberry (2020) and Elenev, Landvoigt, and Nieuwerburgh (2021). Importantly, not only do fluctuations in monetary policy (an aggregate shock) affect bank balance sheets through the bank lending channel but, unlike those papers, idiosyncratic shocks to bank external funding can affect aggregate lending when they hit big (positive measure) banks in what could be called the “granular channel”. In this section, I present models of the heterogeneous banking industry which are validated along the above mentioned bank lending and granularity channels, as well as how endogenous changes in banking market structure may affect financial stability (for lack of a better term, I will call it the competition-(in)stability channel). Much of this section draws on Corbae and D’Erasmus (2021) and Corbae and D’Erasmus (2025), but there is a budding literature building quantitative structural models of the cross-section of banks to address a multitude of policy relevant questions. Some prominent examples include Bianchi and Bigio (2022) and Jamilov and Monacelli (2025); a

³²The model generated dominant Q overestimates the data due to our simplifying assumption of a take-it-or-leave-it merger offer to fringe banks.

³³Further empirical evidence for the bank lending channel can be found in Khwaja and Mian (2008).

distinguishing difference between these models with atomless banks and what I will present below is the absence of granularity and spillovers.

8.1 Lending Channel Environment

The essential elements of the model are guided by the data presented in Section 3:

1. Borrowers with limited liability have a risk-return tradeoff so the endogenous probability of default on a loan depends on interest rates as in Boyd and De Nicolo (2005) owing to selection effects; the process is parameterized to be consistent with chargeoffs on bank balance sheets.
2. Risk neutral households can deposit their earnings in banks, hold bank and non-bank shares yielding dividend flows, inject equity to banks, and pay taxes to cover bank bailouts.
3. There is a dominant-fringe framework as in Gowrisankaran and Holmes (2004), which along with endogenous exit and entry, generates endogenous levels of bank concentration.
4. Banks bear monitoring and fixed costs as in Diamond (1984) consistent with increasing returns to scale we and others find in the data.
5. Idiosyncratic deposit funding inflows are modeled as a finite state, bank size dependent autoregressive process as described in Fact 4 in Section 3 ; bigger banks have lower variance consistent with geographic diversification. This makes bigger banks less prone to exit as in the data.
6. Idiosyncratic shocks to the dominant (granular) banks generate aggregate uncertainty and the possibility of granularity.
7. In Corbae and D’Erasmus (2021), banks can retain earnings, so equity capital is endogenous and banks choose to hold a precautionary buffer that is size dependent as in the data; bigger banks have lower buffers as in a standard precautionary savings model.
8. While big and small banks Cournot compete for borrowers in the bank loan market, borrowers can choose to get a loan from a non-bank as in Buchak et al. (2018). This latter option allows the model to deal with issues of regulatory arbitrage in response to changes in capital requirements or liquidity coverage ratios associated with Dodd-Frank and Basel III.

The essential elements of the model environments in this section bear many similarities with the Merger model in Section 7 given they share the dominant-fringe industry structure. The main differences between the sections are:

- i. There is no merger stage [M1]; only loan stage [M2] and payoff stage [M3].

- ii. Borrowers make a risk-return tradeoff which affects the probability of default, taken as exogenously given in Section 7.1.
- iii. There are persistent idiosyncratic shocks to the supply of deposits taken as transitory in 7.1.
- iv. Banks can retain earnings absent in 7.1 present in Corbae and D’Erasmus (2021).

The addition of persistent idiosyncratic shocks and retained earnings can complicate the state space in much the same way as heterogeneous agent models with aggregate uncertainty, requiring computational methods along the lines of Ifrach and Weintraub (2017).

The dominant fringe framework nests models of perfect competition when entry costs for dominant banks are sufficiently large. The endogenous bank size distribution allows us to validate the model, for instance, against empirical findings of Kashyap and Stein (2000) for the bank lending channel. It also allows us to address issues of how bank regulation affects allocative efficiency in the loan market along the lines of Olley and Pakes (1996).

8.2 Lending Channel Equilibrium

In the interest of space, given that Section 7.2 provided the equilibrium conditions for a related dominant-fringe model with the noted exceptions above, here I will only focus on the key differences in equilibrium conditions and point the interested reader to Corbae and D’Erasmus (2021) and Corbae and D’Erasmus (2025) for a detailed description of equilibrium conditions.

Perhaps most important are the differences in the cross-sectional distribution across papers for a *given aggregate state*. In Section 7, the measure of identical fringe banks with deposits D_f is γ and there is a counting measure defined on $\{0,1\}$ for the dominant bank where 1 and 0 denotes whether the dominant bank is or is not active, respectively, with deposits D_d if active.

In Corbae and D’Erasmus (2025), the cross-sectional distribution is more complicated since there are multiple regions and persistent idiosyncratic deposit shocks. Specifically, in region j for an incumbent bank of type $\theta = \{n, r, f\}$, the type specific distribution μ_θ at any time depends on its individual state s_θ with $s_n = d_n$, $s_r = (j, d_r)$, and $s_f = (j, d_f, c_f^1)$ where $\{n, r, f\}$ denote dominant national, dominant regional, and fringe type banks, d_θ denotes beginning-of-period deposits drawn from a type-dependent Markov Process iid across banks, and c_f^1 denotes a permanent monitoring cost draw for a fringe bank. Note however, that these state variables are exogenous so while the bank size distribution is endogenous due to entry and exit, it is relatively easy to compute allowing us to focus on the strategic game in the loan market between granular banks along the lines of Ericson and Pakes (1995).

While there are no regions in Corbae and D’Erasmus (2021), the cross-sectional distribution at a given time is more complicated since it depends on the endogenous evolution of a bank’s equity capital denoted k . Specifically, for an incumbent bank of type $\theta = \{b, f\}$, the type specific distribution μ_θ at any time depends on its individual state $s_\theta = (k_\theta, d_\theta)$ where $\{b, f\}$ denotes dominant big and fringe type banks and d_θ denotes beginning-of-period deposits drawn from a Markov Process iid across banks consistent with our data analysis by

bank size in Section 3. These cross-sectional distributions are more complicated to compute given the transition functions depend not only on exogenous functions but also endogenous choices due to entry, exit, and dividend payout/retained earnings decisions of the bank.

8.3 Bank Lending Results

After calibrating the model, we use it as a laboratory to run numerous experiments. Here we simply list a few which have previously been discussed.

8.3.1 Model Validation: Bank Lending Channel

Since our focus is on the interaction of policy and market structure for commercial bank lending, it is important that the model be consistent with the bank lending channel. Kashyap and Stein (1995) and Kashyap and Stein (2000) applied a corporate finance approach and argued that if the bank lending channel of monetary policy is correct, one should expect the loan portfolios of large and small banks to respond differently to a contraction in monetary policy. Specifically, is the impact of monetary policy on lending behavior stronger for smaller banks who are more likely to have difficulty substituting into non-deposit sources of external finance? They found strong empirical evidence in favor of this bank lending channel. Their result is driven largely by the smaller banks (those in the bottom 95% of the size distribution). Here we ask the same question of our model.

Corbae and D’Erasmus (2021) implement this policy experiment by analyzing how a permanent rise in the cost of external debt finance (in particular, an increase of 25 basis points in r_D from 0.688% to 0.938%) affects the balance sheet and lending behavior of banks of different sizes. We simulate the model and construct a pseudo-panel of banks under each value of r_D . We then follow a variant of Kashyap and Stein (1995) and estimate the following panel regression by bank size:

$$\Delta\ell_{it} = a_0 + a_1\Delta r_D + a_2X_{it} + u_{it}, \quad (58)$$

where $\Delta\ell_{it}$ denotes the growth rate of loans, Δr_D is the measure of monetary impulse and X_{it} are other bank and aggregate controls. To estimate this regression, we simulate many panels of banks pre- and post-policy change and use the period around the change to estimate the lending response of banks of different sizes. The evaluation of the model response is done around the change in r_D . Table 4 in the Online Appendix presents the estimated coefficients for banks of different sizes.

The results (i.e., a negative and more sizable coefficient on the monetary impulse for small banks than big banks) in Table 4 in the Online Appendix are broadly consistent with Kashyap and Stein’s findings. A contraction in monetary policy does indeed lead to a decline in lending in all size categories of small banks and importantly that the effect is larger the smaller the bank. This result stems from our estimation result that small banks find it harder to raise financing with instruments other than deposits consistent with a pecking order theory.

In Appendix A-4.3 of Corbae and D’Erasmus (2021), we further discuss the market structure and aggregate implications of an unanticipated change to a more contractionary monetary policy. We find that the model exhibits incomplete pass-through, which is consistent with models of imperfect competition such as Drechsler, Savov, and Schnabl (2017), who find that interest spread betas (in their case the spread between the fed funds rate and the deposit rate with respect to changes in the fed funds rate) are less than one, and with Wang et al. (2022), who find that lending spreads decline as the federal funds rate rises. We find that for a 25 basis point increase in r_D , the loan interest rate r_L increases by 16 basis points in the short run, resulting in a 2.21% increase in the interest margin.

8.3.2 Model Validation: Competition-Stability Hypotheses

Here we use our structural model in Corbae and D’Erasmus (2021) to simulate pseudo-data in order to assess its predictions relative to the vast empirical literature on the competition-stability hypothesis. As in Beck, Demircuc-Kunt, and Levine (2006), we estimate a logit model of the probability of a crisis as a function of the degree of banking industry concentration and other relevant aggregate variables. Moreover, as in Berger, Klapper, and Turk-Ariss (2009), we estimate a linear model of the aggregate default frequency as a function of banking industry concentration and other relevant controls. The banking crisis indicator takes value equal to one in periods whenever: (i) the loan default frequency and the exit rate are higher than two standard deviations from their mean; (ii) deposit insurance outlays as a fraction of GDP are higher than 2%; or (iii) large dominant banks are liquidated. The concentration index corresponds to the loan market share of the big bank. We use as extra regressors the growth rate of GDP and lagged growth rate of loan supply. Table 5 in the Online Appendix displays the estimated coefficients and their standard errors.

Consistent with the empirical evidence in Beck, Demircuc-Kunt, and Levine (2006) along with the empirical results in Corbae and Levine (2024) presented in Section 5, we find that banking system concentration is highly significant and negatively related to the probability of a banking crisis. Higher market power induces periods of higher profits that mitigate bank exit. This is in line with the findings of Allen and Gale (2004). Consistent with the evidence in Berger, Klapper, and Turk-Ariss (2009) we find that the relationship between concentration and loan portfolio risk is positive and significant, though not quantitatively large. This is in line with the view of Boyd and De Nicolo (2005), who showed that higher concentration is associated with riskier loan portfolios due to negative selection.

8.3.3 Model Validation: The Marginal Propensity to Lend and Bank Granularity

We study how idiosyncratic (transitory and permanent) bank innovations to deposit flows affect lending and how that translates into bank granularity. To begin, we are interested in understanding the following relationship

$$\Delta L_{it} = \beta_{k(i)} \Delta D_{it} + \zeta_{it} \quad (59)$$

where $\Delta L_{it} = \log(L_{it}) - \log(L_{it-1})$ denotes the growth rate of loans in bank i in period t (or the log-difference) and $\Delta D_{it} = \log(D_{it}) - \log(D_{it-1})$ the growth rate of deposits and $k(i) \in \{d, f\}$ denotes bank type (i.e. whether bank i is a dominant or fringe type), and ζ_{it} is an iid mean zero shock with variance σ_ζ^2 . We will define $\beta_{k(i)}$ as the marginal propensity to loan out of deposit funding (MPL) for bank i of type k .³⁴ One can think of this as an application to banking of the heterogeneous agent, consumption-savings literature studying the marginal propensity to consume (MPC) surveyed in Kaplan and Violante (2022).³⁵

We follow the empirical literature studying the MPC out of income by modeling the deposit shock process with both permanent and transitory components as:

$$\log(D_{it}) = \xi_{k(i)} X_{it} + z_{it} + \varepsilon_{it} \quad \Rightarrow \quad \Delta D_{it} = \xi_{k(i)} \Delta X_{it} + \underbrace{\eta_{it} + \Delta \varepsilon_{it}}_{=\nu_{it}} \quad (60)$$

where $z_{it} = z_{it-1} + \eta_{it}$ and X_{it} can depend on bank characteristics at time $t-1$ as well as other aggregate factors such as the stance of monetary policy. The deposit process in (60) implies that log deposits in each period consists of two iid mean-zero innovations η_{it} and ε_{it} with nonzero bank-type dependent variances $\sigma_{\eta, k(i)}^2$ and $\sigma_{\varepsilon, k(i)}^2$, respectively. The shock ε_{it} is transitory and η_{it} is a permanent innovation.³⁶ We denote the unexplained idiosyncratic variation to deposits in (60) by ν_{it} . Then the deposit process in (60) implies that equation (59) can then be written as

$$\Delta L_{it} = \beta_{k(i)} \xi_{k(i)} \Delta X_{it} + \beta_{k(i)} \nu_{it} + \zeta_{it}. \quad (61)$$

so that the idiosyncratic innovations $\nu_{it} = \eta_{it} + \Delta \varepsilon_{it}$ help identify the MPL $\beta_{k(i)}$. Table 6 in the Online Appendix presents estimates of the marginal propensity to lend from equation (61) in the data. It demonstrates that top 4 banks have a higher MPL ($0.510+0.449=0.959$) for the period 1984-2019 than the smaller fringe (0.510).

Gabaix (2011) constructs size-weighted idiosyncratic shocks to the largest K firms in the economy that forms the basis for his granularity regressions. In the context of our banking problem, there is a connection between the ν_{it} from the MPL regressions and how we implement Gabaix’s granularity results. Specifically, let the granular residual be

$$\Gamma_t^d = \sum_{i=1}^K \omega_{it} \nu_{it} \quad (62)$$

where K denotes the number of “granular” banks, $\omega_{it} = \frac{L_{i,t-1}}{L_{t-1}}$ is the loan market share of bank i , $L_{t-1} = \sum_{i=1}^N L_{i,t-1}$ is aggregate lending, and N denotes the total number of active

³⁴Jamilov and Monacelli (2025) provide an alternative definition of the MPL as the change in loans given a unit change in bank net worth while Gigante (2025) defines it as the change in loans given a unit change in the return on loans. Corell (2025) uses a definition of MPL close to ours.

³⁵Kaplan and Violante (2022) focus on the quarterly MPC out of a \$500 windfall. Importantly, they argue the average MPC of 15% masks substantial heterogeneity. Specifically, many households have MPCs that are close to zero, but some households have MPCs not far from 1, with a great deal of variation in between.

³⁶One may interpret the permanent innovation η_{it} as driven by mergers as in Corbae, D’Erasmus, and Smith (2025) or geographic expansion as in Corbae and D’Erasmus (2020), both of which are endogenous in those models. We interpret ε_{it} as unanticipated deposit shocks along the lines of the process in Corbae and D’Erasmus (2021), Corbae and D’Erasmus (2025), and Corbae, D’Erasmus, and Smith (2025).

banks. Then to answer the question of how much do idiosyncratic innovations to the top $K = 4$ dominant banks explain the variance of aggregate bank lending, as in Gabaix (2011) we estimate the model

$$\Delta L_t = \beta_\Gamma \Gamma_t^d + \epsilon_t^\Gamma \quad (63)$$

and compute the R^2 from the estimated equation (63) given by

$$R^2 = \frac{\beta_\Gamma^2 \text{Var}(\Gamma_t^d)}{\text{Var}(\Delta L_t)}. \quad (64)$$

There are two clear theoretical takeaways from (62) connecting the MPL and granularity results. If each of the K dominant banks were literally of measure zero (i.e. $\omega_{it} = 0$), Gabaix’s R^2 would be zero. Further, if the MPL of the dominant banks were zero, the idiosyncratic shocks would explain none of the variation in aggregate lending captured by Gabaix’s R^2 .

Table 7 in the Online Appendix provides regression estimates of $\beta_\Gamma = 0.474$ in (63) in the data.³⁷ Thus β_Γ is significantly less than the estimated MPL coefficient $\beta_d = 0.959$ in the data in Table 6. In Corbae, D’Erasmus, and Smith (2025) and in equation (69) of the Online Appendix A.3.2, we explicitly show the connection between Gabaix’s results in (63) and the MPL regressions in (61). Further, there we explain why Gabaix’s β_Γ lies below the MPL estimate β_d .

Having estimated MPLs and granularity results in the data, now we use the model in Corbae and D’Erasmus (2025) to estimate the marginal propensity to lend. We find that the estimates align well with certain moments in the data; it is in this way that the results serve as a validation exercise. In particular, the model is calibrated to the period between 1984-2019 and we find that the average MPL for Top 4 banks equals 0.871 in Table 6 in the Online Appendix versus 0.959 in the data. The model average for all other banks not in the Top 4 equals 0.258 in Table 6. While the model average is below the data average of 0.510 in Table 6, the ordering of higher MPL for big versus small banks is consistent between the model and the data.

We can also map our model directly to the approach in Gabaix (2011) to measure the contribution of dominant bank shocks to aggregate fluctuations. We take the estimated finite state, bank size dependent autoregressive process for deposit shocks (truly exogenous variation in the model in Corbae and D’Erasmus (2025) described in Fact 4 in Section 3) to construct the granular residual in (62). Then we use the model generated aggregate loan supply and granular residual in regression (63) to obtain a model consistent R^2 .

The model generated granularity results for how much of the growth in aggregate bank lending can be explained by idiosyncratic shocks to deposit inflows of the largest banks are presented in Table 7 to be contrasted with the analogous data results. We find that the granular residual from model generated data overestimates the variation in aggregate bank credit growth vis-a-vis the data ($R^2 = 0.52$ in the model versus $R^2 = 0.12$ in the data). Using equation (64), the higher R^2 in the model relative to data is attributable to higher β_Γ (i.e. MPL for dominant banks) and lower $\text{Var}(\Delta L_t)$ (since the model sparsely parameterizes uncertainty).

³⁷Here we simply run OLS regressions. See Gabaix and Koijen (2024) for an IV approach.

8.3.4 Regulatory Responses to the Financial Crisis

After the financial crisis of 2008, efforts to increase financial stability via capital and liquidity requirements have been coordinated by the Basel Committee on Banking Supervision at the Bank for International Settlements (BIS) in what is referred to as Basel III rules. Since our focus is on the interaction between regulatory policy aimed towards stability and market structure, in Corbae and D’Erasmus (2021) we ask the question, how much does an increase from 4% to 8.5% in capital requirements affect bank exit, market shares of big and small banks, and other outcomes? Given our focus on market structure we consider the policy change in both our baseline model of imperfect competition and a model of perfect competition in the loan market calibrated to match our baseline data moments. Table 8 in the Online Appendix presents short and long run results for the imperfect competition case.³⁸

We start by describing the long-run effects of the policy change in the model with imperfect competition. As columns (2) and (5) in Table 8 show, we find that an increase in capital requirements from 4% to 8.5% leads to a 20% decline in long run exit (an intended consequence) and entry rates of small banks and decline in small bank loan market share of 6% leading to a more concentrated industry (an unintended consequence). The sizable change in big bank market share is explained by an extensive margin drop of 15% in the number (measure) of fringe banks as well as the larger increase in the intensive margin by big banks than that of surviving fringe banks (4% versus 0.7%). The net effect of higher big bank lending and lower small bank lending is a decrease in total bank lending of 9%. This leads to a 6 basis points increase in interest rates and a 1.4% increase in markups on bank loans (which benefits not only big banks with market power but also the fringe banks). The meager increase in bank interest rates is due to the shift towards non-bank lending so that total credit declines by only 0.3% and output decreases by only 0.2%. This provides an example of regulatory arbitrage along the lines of Buchak et al. (2018); when bank capital requirements are increased borrowing shifts to the nonbank sector. Due to selection effects, the decline in exit rates together with the fact that exiting banks are better capitalized and have a smaller share of assets invested in loans results in a nearly 70% decline in the taxes-to-output ratio (which is relevant for welfare effects we present later).

The most significant difference between the short run impact and the long run effect of the policy (i.e. columns 4 and 5 in Table 8) is the increase in exit rates (37%) and the collapse of bank entry (a decline of 96%). The decline in bank entry is consistent with the data as there has been no denovo entry (i.e., newly created banks) between 2012 and 2016.³⁹ The resulting increase in loan market concentration (a 5% reduction in fringe market share in the short-run) leads to a significant increase in markups (a competitive effect together with a composition effect) in the short run. Specifically, the big bank increases its market share in response to the market share left by failing banks (a selection effect) and the reduction in lending by those fringe banks that remain active. This distributional change contributes to a reduction in average costs as well as an increase in price, leading to the increased markup.

Of particular importance for financial stability is the change in the long run probability of

³⁸See Table X in Corbae and D’Erasmus (2021) for the comparison to a calibrated competitive model.

³⁹Liu (2025) considers the effect of regulatory changes on de novo entry by fringe banks.

a crisis as a result of the policy. The policy change in the model with imperfect competition results in a decline of 20% in the probability of a crisis as the big bank gains market share, the reduction in the exit rate is accompanied by a decline in its volatility. Corbae and D’Erasmus (2021) show that there are actually qualitative differences between the financial stability implications of raising capital requirements in a model with imperfect competition than one with perfect competition in Table 10 of their paper. In particular, the probability of a crisis actually rises in the model with perfect competition. In summary, bank regulation can have a significantly different response in the model with imperfect competition than in a model with perfect competition.

8.3.5 How Does Regulation Affect Allocative Efficiency?

An important aspect of the policy reforms we study in Corbae and D’Erasmus (2021) is that they may change the level of allocative efficiency in the economy by shifting lending between heterogeneous banks. In order to explain this change and to provide a measure that captures how efficiently lending is allocated in the economy, we use the following decomposition of weighted average bank-level marginal cost (proposed originally by Olley and Pakes (1996) to measure productivity):

$$\widehat{c}(z) \equiv \sum_{\theta} \int c_{\theta}(\ell'_{\theta}, z) \omega(\ell'_{\theta}) d\mu_{\theta}(k_{\theta}, \delta_{\theta}) = \bar{c}(z) + cov(c_{\theta}(\ell'_{\theta}, z), \omega(\ell'_{\theta})), \quad (65)$$

where $\widehat{c}(z)$ is the loan-weighted average of bank-level cost in a period when aggregate productivity is z , $c_{\theta}(\ell'_{\theta}, z)$ is the net cost of extending ℓ'_{θ} , $\omega(\ell'_{\theta})$ is the loan share, and $\bar{c}(z)$ is the un-weighted mean cost (i.e., $\sum_{\theta} \int c_{\theta}(\ell'_{\theta}, z) d\mu_{\theta}$).⁴⁰ That is, loan weighted average cost can be decomposed into two terms: the un-weighted average of bank-level cost and a covariance term between loan shares and cost. A smaller value for the covariance term captures an improvement in allocative efficiency (since the distribution of loans shifts towards banks with lower costs). Since we documented differences in costs $c_{\theta}(\cdot)$ between big and fringe banks in point (5) of the Data Section 3, allocative efficiency can improve as loans are reallocated from high cost fringe banks to big banks with declining average costs (increasing returns over the region we have estimated).

Since access to cheap external finance is capacity constrained by $d_{\theta}(z)$, we also present a decomposition of average (loan-weighted) deposits $\widehat{d}(z) = \bar{d}(z) + cov(d_{\theta}(z), \omega(\ell'_{\theta}))$. This decomposition shows how the distribution of lending and access to external financing (which affects the cost of financing loans) change with regulatory policy. For a given level of aggregate lending, as banks with a higher d_{θ} take over a larger portion of the market, access to cheap external financing increases. For this reason, a larger value for the covariance term $cov(d_{\theta}(z), \omega(\ell'_{\theta}))$ captures an improvement in allocative efficiency (since the distribution of loans shifts towards banks with higher d_{θ}). We view allocative efficiency as measured by $cov(c, \omega)$ as capturing costs on the asset side and $cov(d, \omega)$ as capturing costs on the liabilities side of the bank balance sheet.

⁴⁰Like the long-run moments presented in the previous section, we compute the average (loan-weighted) cost in equation (65) and all its components every period during the simulation of the model. Table 9 presents the corresponding time series average.

Table 9 in the Online Appendix shows the values for the two decompositions in the baseline model with imperfect competition and a (recalibrated) model of perfect competition. Starting with a comparison of the average cost results in the baseline parameterization, we observe that average (unweighted) costs are almost identical between the model with imperfect competition (column 2) and perfect competition (column 3) but allocative efficiency as measured by $cov(c, \omega)$ is significantly higher (i.e., the covariance is more negative) in the model with imperfect competition. This better allocation of lending activity is reflected in a significantly lower measured loan weighted average cost $\hat{c}$ in the model with imperfect competition than in the model with perfect competition as a significant fraction of loans are extended by big banks, which have lower average costs than fringe banks. Allocative efficiency as measured by $cov(\delta, \omega)$ is also better (more positive) in the model with imperfect competition than the perfectly competitive model, again due to distributional differences.

Turning to the effect of higher capital requirements, they result in an increase in allocative efficiency (as measured by the average (loan-weighted) costs) but the increase is quantitatively more significant in the model with imperfect competition (column 2 versus column 4) as opposed to the model with perfect competition (column 3 versus column 5). The differences arise as the change in average costs comes not only from changes within the distribution of fringe banks but also from changes in loan market concentration. In addition, we observe that the differences in bank industry dynamics described in the previous section also have important implications for the distribution of deposits d . In the model with imperfect competition, the increase in concentration and the reduction in bank failure after the increase in capital requirements are large enough to induce an increase in average (loan-weighted) deposits $\hat{d}(z)$ that is explained completely by an increase in allocative efficiency as measured by $cov(d, \omega)$ (as in fact average deposits $\bar{d}(z)$ decline). On the other hand, the significantly smaller decline in bank exit (and the absence of changes in big bank market share) in the perfectly competitive case (column 3 versus column 5) leads to a slight decline in $\hat{d}(z)$ that is explained by a decline in allocative efficiency as $cov(d, \omega)$ decreases (as $\bar{d}$ increases).

In sum, after all policy changes, the distribution of loans shifts towards banks with lower costs both in monitoring loans and in obtaining cheap external finance, which drives the improvement in allocative efficiency in our baseline model of imperfect competition (unlike the perfectly competitive model). This relationship between banking regulation and allocative efficiency is consistent with the findings of Berger and Hannan (1998) that present evidence in favor of efficiency gains from an increase in bank concentration.

8.3.6 Too-Big-To-Fail

To quote Ben Bernanke “I want to be very, very clear: too-big-to-fail is one of the biggest problems we face in this country, and we must take action to eliminate too-big-to-fail.”⁴¹ The top 4 commercial bank loan market share was 40% in the last quarter of 2007. Since our framework admits a nontrivial endogenous size distribution of banks and includes big, granular banks, it is well suited to analyze size dependent policy counterfactuals like that

⁴¹Ben Bernanke, Time, December 28, 2009/January 4, 2010, p. 78.

described above by Ben Bernanke.

In the baseline economy of Corbae and D’Erasmus (2025), we consider the possibility of national bank bailouts in states where national banks experience negative profits which, without a sufficiently high seasoned equity injection, would lead to bank failure. A policy of big bank bailouts or “too-big-to-fail” (TBTF) implies that with some probability (95% in our calibrated economy) the government will bail out national banks in the event of realized losses. Such a policy changes the ex-ante incentives of national banks since they can take on more risk if they are guaranteed that they receive ex-post bailouts.

To assess the role of TBTF on risk taking, we compare our baseline economy with a counterfactual economy where there are no government bailouts to national banks.⁴² As before, the bank will optimally decide to stay in operation or to exit after observing the realization of profits. Since the charter value of a bank is always non-negative due to limited liability, as the government covers a smaller fraction of the loss in this counterfactual, the probability of the bank continuing in operation decreases. In Table 10 in the online Appendix, we present the comparison of our calibrated baseline to the case with no bailouts. In our benchmark equilibrium with national bank bailouts there was no national bank failure on-the-equilibrium path. We find that after removing national bank bailouts, failure doesn’t arise on-the-equilibrium path because national banks reduce their loan exposure in order to maintain their charter value. We now observe a decline of 2.6% in the probability of a national bank reaching a state with negative profits. There is a significant decline in taxes to output (-28%) as the government eliminates the bailout policy.

As evident in Table 10 in the online Appendix, eliminating TBTF induces the national bank to reduce its loan supply and a portion of the market is taken over by regional and fringe banks (their market share increases by 1.5 and 1.6 percent, respectively). The increase in the loan supply of smaller banks is not enough to compensate for the decline in lending by the national bank resulting in a decline in total bank lending (-0.75%) which in turn leads to higher interest rates (+0.32%) and default frequency (+0.19%). The higher default frequency is compensated by higher interest rates resulting in higher profits along the equilibrium path (markups increase by 0.98%) which leads to a decline in failure rates (-0.35%). The decline in lending induces a reduction in output by 0.07% but the reduction in failures does not translate into a reduction in the ratio of taxes for failure to output.

To assess the welfare consequences of changes in bailout policies, we ask the question, “What would households and entrepreneurs be willing to pay (or be paid) to remove national bank subsidies?” To answer this question, we calculate consumption equivalents for each type of representative agent household (H) and entrepreneur (E). Specifically, let $C_{h,t}^{pre}$ denote type $h \in \{H, E\}$ equilibrium aggregate consumption in the baseline economy with TBTF and $C_{h,t}^{post}$ denote type h equilibrium aggregate consumption post-reform without TBTF. Then, assuming the consumption flow of the entrepreneurs is discounted at the same rate as the households, the consumption equivalent α_h for a representative type h agent as a result of the reforms is defined as the constant percentage increase in consumption in the pre-reform case that gives the agent the same expected utility in the post-reform economy. Specifically,

⁴²Other important papers that study the TBTF problem and bank resolution policies are Dávila and Walther (2020), Allen et al. (2023), and Meehl (2025).

α_h solves:

$$E_0 \left[\sum_{t=0}^{\infty} \beta_h^t C_{h,t}^{post} \right] = E_0 \left[\sum_{t=0}^{\infty} \beta_h^t (1 + \alpha_h) C_{h,t}^{pre} \right]. \quad (66)$$

We let the ex-ante value (i.e., before being born as $h \in \{H, E\}$) for an agent in the pre-reform economy be

$$W_{pre} = \frac{\mathcal{H}}{1 + \mathcal{H}} E_0 \left[\sum_{t=0}^{\infty} \beta_H^t C_{H,t}^{pre} \right] + \frac{1}{1 + \mathcal{H}} E_0 \left[\sum_{t=0}^{\infty} \beta_E^t C_{E,t}^{pre} \right]. \quad (67)$$

where $\mathcal{H}$ denotes the mass of households in the economy.⁴³ We define the ex-ante value post-reform similarly. Then, we can write the ex-ante value of the reform in consumption equivalent terms $\bar{\alpha}$ as $W_{post} = (1 + \bar{\alpha})W_{pre}$. Equations (66) and (67) make clear that, since preferences are linear, computing ex-ante consumption equivalents in the long-run amounts to computing the difference between expected consumption pre- and post-reform.

Column (2) in Table 11 in the Online Appendix shows that both households and entrepreneurs would be willing to pay in order to remain in the economy with national bank bailouts as $\alpha_E = -0.045\%$ and $\alpha_H = -0.002\%$. In the case of entrepreneurs, the decline in welfare (measured in consumption equivalents) derives from the decline in lending. In the case of the households, while in the baseline the households face taxes to cover deposit insurance, after removing the bailout policy, as the owners of the banks, the representative household is required to still inject equity to cover negative profits plus equity issuance costs so the change in taxes almost cancels out. We calibrate $\mathcal{H} = 4.88$ as in Corbae and D’Erasmus (2021) so the fraction of entrepreneurs equals 17%. This leads to an aggregate welfare loss (in consumption equivalents) of 0.006%.

Since we assume households and entrepreneurs are risk neutral, this normative result abstracts from changes in aggregate second moments. Table 11 also includes a calculation of the variance of gdp and consumption in the absence of TBTF. Without TBTF, big banks internalize that risky lending raises the likelihood of failure and loss of charter value. Concern about the loss of charter value induces banks to take less risk which leads to a 3% lower variance of lending. A lower variance of lending induces a drop in the variance of gdp. While this leads to lower variance of consumption for the entrepreneur, household consumption variability actually rises because they need to inject equity to cover negative profits adding to variability of household cash flows. In our parameterization, we find an increase in the variability of household consumption results in an increase in the variability of aggregate consumption. Thus, our estimate yields a lower bound on welfare losses because TBTF policies actually generate lower variance of aggregate consumption.

9 Directions for Future Research

Here I suggest some directions for future research that would be useful to be addressed in quantitative models.

⁴³Since there is a mass $\mathcal{H}$ of households and a unit mass of entrepreneurs the probability of being born as a household is $\frac{\mathcal{H}}{1+\mathcal{H}}$.

- **Economies of Scope:** While much of the previous sections have focused on how relaxation of cross-state branching restrictions in Riegle-Neal allowed banks to exploit economies of scale and the benefits of diversification of idiosyncratic shocks to nondifferentiated loans and deposits, little focus was placed on economics that of scope that were made possible by the Gramm–Leach–Bliley Act of 1999. Economies of scope can also generate diversification by offering differentiated products.⁴⁴ Economies of scope can also lead to a reduction in the average costs when a bank offers a variety of related products since shared resources, infrastructure, or knowledge are used to offer different products, reducing the need for separate investments and operations.⁴⁵ Another dimension along these lines is choice of business models targeted at different clienteles. d’Avernas et al. (2024) study how different size banks target different depositor clienteles.
- **Complexity:** While Section 5 focused on misaligned preferences between managers and shareholders, complexity has become a more important issue as banks grow larger and their span of products grow in accord with the Gramm–Leach–Bliley Act of 1999. Growing complexity can be an important issue for systemic risk, and is taken up empirically in Correa and Goldberg (2022) by constructing measures of complexity and testing how they affect risk taking. Future research using quantitative models to understand the tradeoffs in bank opacity along the lines of the simple 3 period model in Dang et al. (2017) to governance and supervision by regulators. Some deep theory on information design and private information can be addressed in this literature.
- **Global Banking:** A related under-researched quantitative structural agenda which shares the objectives of understanding geographic expansion and complexity is that of global banking. As stated (p.167) in Buch and Goldberg (2020) “A second lesson is that the response of global banks to shocks is highly heterogeneous...Responses to shocks are larger when bank-specific capabilities to absorb shocks are weaker, when bank capital is lower, and when bank internal liquidity management is less effective. Responses also depend on the business model of the parent bank, the importance of the foreign market for the overall balance sheet, and the degree of its geographic or organizational complexity. Country specific macroeconomic factors matter as well, including the policy environment...This is the point at which micro- and macroeconomic views need to be combined.”

Further, Buch and Goldberg focus on global equivalents of the costs and benefits of expansion that we have elucidated in earlier sections following relaxation of state

⁴⁴For instance, the fact that banks offer both term loans (intertemporal smoothing) and credit lines (which can be thought of as an insurance product) are subject to imperfectly correlated shocks. See Choi (2025) for such a model.

⁴⁵In Corbae and D’Erasmus (2021) and Corbae and D’Erasmus (2025) lower average costs for large banks arise from both channels. Specifically, non-interest marginal net costs capture non-interest income/expenses from other products (e.g. income from fees on credit lines as discussed above). While those papers only account for economies of scope in this exogenous way, Corbae and D’Erasmus (2020) endogenize the choice of technology using a quality ladder approach to exploit economies of scope and scale.

branching in the U.S. Specifically, (p.142) “Global activities of banks can contribute to the allocation of capital to sectors and regions where rates of return are highest, thus improving the efficiency of production and providing savers with investment opportunities. Cross-border holdings of asset portfolios can improve the diversification of risks and thus, overall, reduce the volatility of consumption. Global banks also may have advantages in accessing external capital markets, overcoming in part some of the frictions facing local banks, and can likewise potentially provide credit on more favorable terms. Heightened competitive pressure can spur local firms to use resources more effectively.”

These benefits, however, bring with them costs which became apparent in the Global Financial Crisis of 2008. Specifically, as the authors state (p.146) “The crisis painfully demonstrated the dark side of banking globalization: Global expansion strategies of banks often did not take due account of the risks involved; global banking organizations reached a degree of complexity that raised the costs of restructuring and resolving failing entities, especially in cross-jurisdictional settings; cross-border expansion was driven not only by the objective of better servicing customers and the real economy but also by regulatory and tax arbitrage; and some activities led to shock amplification rather than mitigation.” An example of a structural model that evaluates these costs and benefits of global banking is Corbae and D’Erasmus (2015).

- Optimal Regulation: Once we have a well parameterized laboratory to conduct counterfactuals, the next step is to compute optimal regulatory policies. While this has been done in several papers listed above, many of the models take the market structure as exogenously given. It is critical to evaluate how regulation affects market structure in the long run which has implications for welfare and allocational efficiency. Since the literature cited above is for a decentralized economy, it would be useful to compare those results to a planning problem with similar frictions to assess how far we are away from a constrained efficient allocation.

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A Online Appendix

A.1 Tables

Table 1: Balance Sheet by Bank Size

	Bank Size			
	All Banks	Top 4	Top 5-35	Fringe
Assets (% of Total Assets)				
Securities (a^+)	35.68	30.43	30.81	35.70
Loans (ℓ)	53.83	61.30	59.96	53.80
Fixed & Intangible Assets (Υ)	10.49	8.27	9.22	10.50
Liabilities (% of Total Assets)				
Deposits (d)	88.38	83.80	83.25	88.41
Fed Funds (a^-)	1.14	7.93	7.53	1.11
Equity (e)	10.49	8.27	9.22	10.50

Notes: From Table 3 in Corbae and D’Erasmus (2025) for 1984 to 2019. Securities include cash plus federal funds sold plus securities. Loans include all loans on the balance sheet and fixed and intangible assets includes fixed assets and other assets not classified into securities or loans (except trading assets). Deposits include total deposits and other borrowings (which include home loan advances). Fed Funds refer to federal funds purchased. This simple categorization accounts for approximately 96% of total assets and about 95% of total liabilities from the Call Report data. We normalize the categories so the data totals add up to one in Table 1. Source: Reports of Condition and Income (Call Reports).

Table 2: On Competition, Governance, and Bank Risk

	Model Prediction	Regression Coefficient
Bank Competition	$\gamma_C > 0$	0.5994*** (0.1778)
Leverage-Lagged	$\gamma_L > 0$	0.0119** (0.0048)
Ln(Bank Assets)-Lagged	$\gamma_A < 0$	-0.1919** (0.0748)
% Institutional Ownership	$\gamma_G < 0$	-1.1725*** (0.1968)
Leverage*Institutional Ownership	$\gamma_I > 0$	0.0497*** (0.0129)
Observations		1994
R-squared		0.7945

Source: Corbae and Levine (2024). Notes: The sample consists of BHC-year observations from 1987 through 1995. *, **, and *** indicate significant at 10%, 5%, and 1%, respectively.

Table 3: Model With and Without Merger Restrictions

Moment	Calc	Pre-RN	Post-RN Transition	Post-RN Long-run
Top 4 Deposit Market Share	$\frac{D_d}{D_d + \gamma D_f}$	13.90%	35.62%	42.34%
Top 4 Loan Market Share	$\frac{L_d}{L_d + \gamma L_f}$	17.25%	32.54%	35.42%
Bank Loans to Total Loans Ratio	$\frac{L_d + \gamma L_f}{L_n + L_d + \gamma L_f}$	42.05%	41.18%	40.33%
Interest Rate	r_B^L	4.73%	4.81%	4.89%
Top 4 Loans/Deposits	$\frac{L_d}{D_d}$	99.28%	69.28%	63.58%
Fringe Loans/Deposits	$\frac{L_f}{D_f}$	77.62%	81.01%	85.65%
Fringe Bank Failure Rate	$\frac{\gamma x}{\Gamma}$	1.23%	1.09%	1.15%
Merger Rate	$\frac{\gamma - \Gamma}{\Gamma}$	0.0%	2.21%	0.02%
Top 4 Loan Markup	$\frac{\theta r_B^L}{r_D + c'_d(L_d)}$	4.43	4.51	4.60
Fringe Loan Markup	$\frac{\theta r_B^L}{r_D + c'_f(L_f)}$	1.49	1.47	1.43
Top 4 Tobin's Q	$\frac{V_d}{K_d}$	1.001	15.32	14.97
Fringe Tobin's Q	$\frac{V_f}{K_f}$	0.61	0.91	0.98

Note: Model “Pre-RN” constrains the dominant bank from merging. Model “Post-RN Transition” removes the pre-Riegle-Neal merger restriction and computes the values along the short-run transition path.

“Post-RN Long-run” presents the values in the long-run, stochastic steady state. Post-RN dominant Tobin's Q uses $K'_d = K_d + (\gamma - \Gamma)K_f$ to capture changes in capital acquisitions during merger activity.

Table 4: Kashyap and Stein ('95) Regressions (Model Pseudo-Panel)

	Dependent Variable $\Delta \ell_{it}$
Specification	Coeff. on Monetary Impulse (Δr^s)
Small 98%	-0.180
s.e.	0.036
Small 92%	-0.265
s.e.	0.036
Small 68%	-0.356
s.e.	0.048

Note: Regressions include one lag of the dependent variable, and growth rate of GDP. All coefficients significant at 1% level.

Table 5: Competition and Stability

Model	Logit	Linear
Dependent Variable	Crisis _{t+1}	Default Freq. _{t+1}
Concentration _t	-27.224 (2.018)	0.057 (0.000)
GDP growth in t	178.80 (37.63)	0.051 (0.029)
Loan Supply Growth _t	1.992 (5.106)	-0.0163 (0.008)
R^2	0.69	0.47

Note: Standard Errors in parentheses. R^2 refers to Pseudo R^2 in the logit model. “Concentration” coefficients significant at 1%.

Table 6: Marginal Propensity to Lend (Data vs Model Estimates)

	Dependent Variable $\Delta \log(L_{it})$	
	Data	Model
ν_{it}	0.510*** (0.00101)	0.2576*** (0.0054)
$\nu_{it} \times I_{t4}$	0.449*** (0.0359)	0.6129*** (0.0062)
Bank FE	✓	
Time FE	✓	
Bank Controls	✓	
N	946299	3872061
R-sq	0.362	0.913
adj. R-sq	0.348	0.912

Note: Standard Errors in parenthesis. *** significant at 1%, ** significant at 5%, * significant at 10% level. Table presents the estimates from equation (59). L_{it} refers to loans and leases. ν_{it} corresponds to the error term as defined in equation (60). We follow Gabaix (2011) and construct ν_{it} by removing a time and bank fixed effect. $I_{t,k(i)=4}$ is an indicator that takes value 1 if bank i is in the Top 4 of the asset distribution. Bank controls includes log-assets, equity/assets, loans/assets, return on assets, salaries/assets, expenses on fixed assets/assets, cost of deposits. Source: Call Reports

Table 7: Granular Banks and the Marginal Propensity to Lend (Data vs Model Estimates)

	Dependent Variable		
	Data	Model	Data
	ΔL_t	ΔL_t	$\sum_{i=1}^N \omega_{it} \Delta L_{it}$
Γ_t^d	0.517*** (0.141)	0.511*** (0.011)	0.968*** (0.170)
constant	0.0081*** (0.001)	0.0004 (0.000)	0.0141*** (0.001)
N	144	1899	144
R^2	0.087	0.5212	0.185
Adjusted R^2	0.080	0.5209	0.180

Note: Standard Errors in parenthesis. *** significant at 1%, ** significant at 5%, * significant at 10% level.
Source: Call Reports.

Table 8: Benchmark Model (Imperfect Competition)

(φ_f, φ_b)	Baseline (0.04, 0.04)	High Capital Requirements (0.085, 0.085)	
		Short Run Δ (%)	Long Run Δ (%)
Moment	(%)		
Capital Ratio (b, f)	(10.24, 10.89)	(37.56, 35.80)	(44.24, 53.65)
Exit Rate	0.87	37.09	-20.20
Entry Rate	0.90	-96.32	-20.32
Prob. of Crisis	0.14	-	-20.00
Loan mkt sh. f	70.81	-4.85	-6.07
$L^{s,c}$ /Total Credit	53.40	-7.11	-8.66
Loan Int. Rate r^c	4.67	0.95	1.17
Avg. Markup	95.71	12.59	1.37
<i>Additional Moments</i>			
Measure f Banks		-3.92	-15.19
Loans (b, f)		(3.65, -7.98)	(4.45, 0.73)
Bank Loan Supply		-7.19	-8.90
Total Credit		-0.08	-0.27
Output		0.00	-0.19
Taxes/Output		-30.02	-68.50
Column	(1)	(2)	(3)

Note: $\Delta(\%)$ refers to the percentage change relative to the baseline model with capital requirements at $\varphi_\theta = 0.04$ (columns (2) and (3)) for the corresponding model with imperfect competition or with perfect competition. The long-run moments are computed from 10,000 period simulations that discard the initial 2,000 periods. Initial conditions for short-run simulations are consistent with the long-run average distribution of banks (big and fringe) as well as aggregate productivity. The moments reported as the “short-run” effects correspond to the moments that arise five periods after the policy change.

Table 9: Allocative Efficiency of Capital and Liquidity Requirements

	Baseline		High Cap. Req.	
(φ_f, φ_b)	(0.04, 0.04)		(0.085, 0.085)	
	Imperfect	Perfect	Imperfect	Perfect
Moment (%)	Comp.	Comp.	Comp.	Comp.
Avg. (loan-weighted) cost $\hat{c}$	1.718	1.802	1.719	1.824
Avg. cost $\bar{c}$	1.759	1.759	1.807	1.791
$Cov(c, \omega)$	-0.041	0.043	-0.089	0.033
Avg. (loan-weighted) deposit $\hat{\delta}$	0.229	0.271	0.241	0.271
Avg. deposit $\bar{\delta}$	0.202	0.261	0.198	0.264
$Cov(d, \omega)$	0.027	0.010	0.043	0.007
Fringe Loan Market Share	70.810	100.000	66.506	100.000
Column	(2)	(3)	(4)	(5)

Note: Moments presented here correspond to time series averages over z .

Table 10: Baseline vs. No Bailouts

	Baseline National Bank Bailout	No Bailouts Change (%)
	(1)	(2)
Bank Loan Supply	0.24	-0.75
Loan Market Share Fringe (%)	25.36	1.63
Loan Market Share Regional (%)	37.18	1.46
Loan Market Share National (%)	37.46	-2.55
Bank Loan Interest Rate (%)	4.32	0.32
Markup (%)	176.24	0.98
Default Frequency (%)	2.23	0.19
Failure Rate (%)	0.95	-0.35
Prob. $\pi_n < 0$	5.69	-2.59
Taxes/Output (%)	0.062	-28.22

Note: Change (%) corresponds to the percentage change between the corresponding model and the Baseline with National Bank Bailouts.

Table 11: Large Bank Bailouts: Normative Results

	Baseline National Bank Bailout	No Bailouts Change (%)
	(1)	(2)
Consumption Entrep	1.028	-0.045
Consumption Households	2.006	-0.002
Agg Consumption	1.839	-0.006
Var(Agg Consumption)	0.0080	0.666

Note: Change (%) corresponds to the percentage change between the corresponding model and the Baseline with National Bank Bailouts. “Agg Consumption” refers to aggregate consumption, Consumption Entrep to consumption entrepreneurs.

A.2 Figures

Figure 3: Governance and Regulatory Counterfactuals

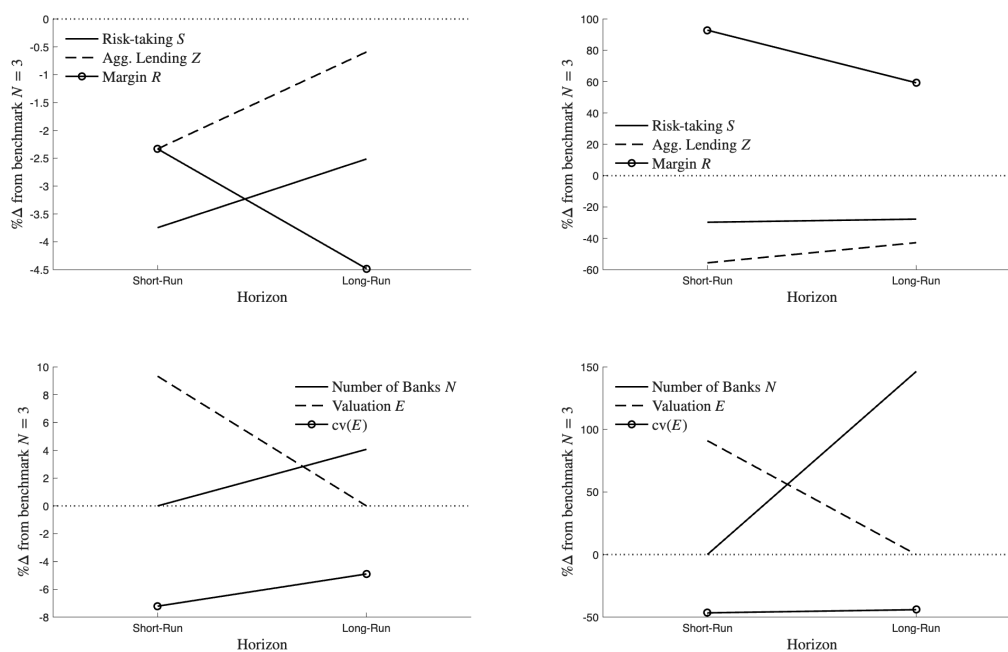
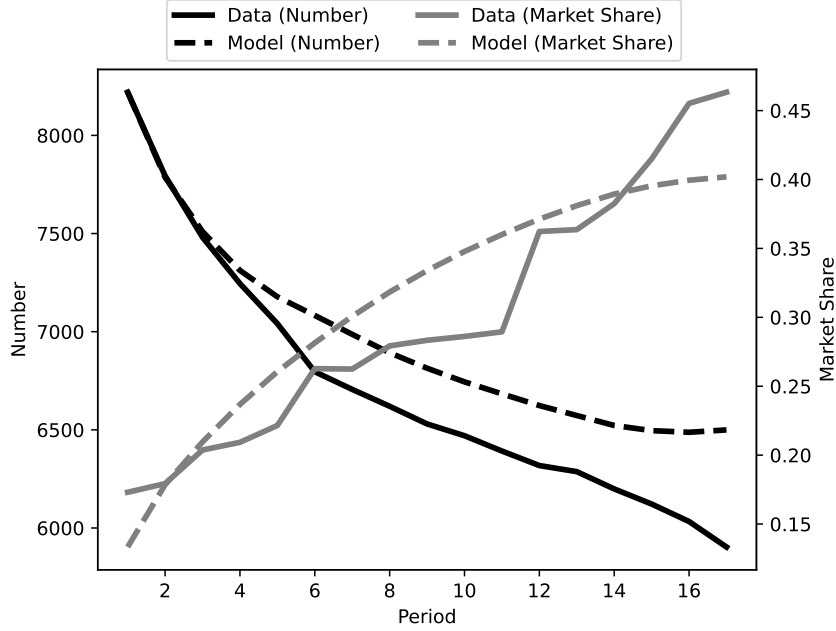


Figure 4: Model vs Data Transition Path (1994 - 2011)



Notes: From Corbae, D’Erasmus, and Smith (2025). Data Number of banks and Top Ten Market Share correspond to Figure 1 from Call Reports. Model number of fringe banks normalized to match the number in the data at start of the period ($\frac{\gamma \cdot 3300}{1000}$).

A.3 Further Results

A.3.1 Monetary Policy Counterfactual

To understand the role of competition for bank stability and the monetary transmission mechanism, we can use the model to analyze the impact of a policy that increases the marginal cost of funds α by 50 basis points (from 0.03 in the benchmark to 0.035). One way to interpret this is a contractionary monetary policy (i.e., a rise in the Fed Funds rate). In the short run (i.e. with $N = 3$ fixed at the benchmark), a rise in the marginal cost of funds leads banks to take on more risk resulting in a lower success probability. Contractionary monetary policy has the intended consequence of lowering aggregate lending/investment Z in the short run at the expense of bank profitability π and equity values E . and despite lower individual bank lending D , the market value of leverage D/E rises since E falls more than D . Importantly, equity volatility rises in the short run with contractionary monetary policy.

This counterfactual illustrates an important observation for regulation. In the short run, leverage regulation based on historical book value (i.e. D/κ) falls since book equity κ is evaluated at the time of entry before the policy shock arises, while the market value of equity as discussed above rises. In the long run, however, book and market value of leverage are rebalanced so that $E = \kappa$. Thus, our counterfactual illustrates that short run risk taking

can rise along with market value of leverage at the same time that book value of leverage falls. While our counterfactual is simply illustrative, the short run policy implications of book versus market valuations are studied in more detail in Begenau et al. (2026).

Since profitability and equity values fall in the short run, the incentive for bank entry also falls thereby leading to less long run competition. Falling competition, which *ceteris paribus* leads banks to take less risk, offsets the short-run increase in risk taking so that there is a zero long-run impact on risk taking (relative to the benchmark). This translates to zero long run effect on volatility of equity. Thus, a contractionary monetary policy, for instance, could lead to short run instability but not affect long run stability. Further, the short-run decrease in aggregate lending is magnified in the long-run since there are fewer banks.

We can also use the model to consider the impact of contractionary monetary policy across different market structures. In particular, we ask what is the effect of increasing α by 50 basis points to 0.035 from 0.03 in the benchmark ($N = 3$) market structure versus a less competitive ($N = 2$) economy on bank risk taking and lending. The former is relevant for thinking about the interaction of financial stability and monetary policy and the latter is relevant for thinking about the monetary transmission mechanism studied in papers such as Kashyap and Stein (2000). Both answers effectively rest upon difference-in-difference studies of contractionary policy across banks of different degrees of sizes or levels of liquidity (i.e. distance from their leverage constraint). Along the first dimension about the interaction of monetary policy and competition on bank risk taking, in Corbae and Levine (2024) we show that the difference-in-difference cross partial calculation $\frac{\partial^2 S}{\partial N \partial \alpha} < 0$. Thus the model predicts a negative interaction effect of contractionary policy on risk taking as competition increases. Further we show that increased competition drives banks closer to their leverage constraints and that leverage constrained banks decrease their lending more in response to a rise in external funding costs induced by contractionary monetary policy. The fact that more competition among smaller banks can have a significant impact on bank lending channel is consistent with the findings in Kashyap and Stein (2000).

As with the previous empirical work on governance, we can also assess the implications of competition on the impact of monetary policy on bank risk taking using the following regression specification:

$$Y_{bst} = \gamma_C \cdot C_{bst} + \gamma_L L_{bst-1} + \gamma_A A_{bst-1} + \gamma_F FFR_{bst} + \gamma_I FFR_{bst} \cdot C_{bst} + \theta_b + \theta_{st} + \varepsilon_{bst} \quad (68)$$

where our measure of the bank specific Fed Funds rate FFR_{bst} is intended to measure sensitivity of banks with less access to non-deposit finance to fed funds rate innovations (specifically, $FFR_t * [(bank\ liabilities - non-deposit\ liabilities)/bank\ liabilities]_{t-1}$).

Table 12: On Competition, Monetary Policy, and Bank Risk

	Model Prediction	Regression Coefficient
Bank Competition	$\gamma_C > 0$	0.5963*** (0.2716)
Leverage-Lagged	$\gamma_L > 0$	0.0297** (0.0049)
Ln(Bank Assets)-Lagged	$\gamma_A < 0$	-0.1580* (0.0897)
% Fed Funds Exposure	$\gamma_F > 0$	1.3956*** (0.4059)
FFR*Competition	$\gamma_I < 0$	-0.4701*** (0.1614)
Observations		1518
R-squared		0.8188

Source: Corbae and Levine (2024). Notes: The sample consists of BHC-year observations from 1987 through 1995. *, **, and *** indicate significant at 10%, 5%, and 1%, respectively.

A.3.2 The Connection Between MPL and Granularity

To see the connection between the Gabaix's results and the MPL regressions, after multiplying both sides of equation (61) by ω_{it} and summing over i while assuming the weighted average growth rate of loans $\sum_{i=1}^N \omega_{it} \Delta L_{it} \equiv \sum_{i=1}^N \omega_{it} (\log(L_{it}) - \log(L_{it-1}))$ is a good approximation of the growth rate of aggregate loans $\Delta L_t \equiv \log(\sum_{i=1}^N L_{it}) - \log(\sum_{i=1}^N L_{it-1})$, equation (63) can be written

$$\Delta L_t \approx \sum_{i=1}^N \omega_{it} \Delta L_{it} = \beta_d \underbrace{\sum_{i=1}^K \omega_{it} \nu_{it}}_{\Gamma_t^d} + \underbrace{\beta_f \sum_{i=K+1}^N \omega_{it} \nu_{it} + \sum_{i=1}^N \omega_{it} (\beta_k \xi_k X_{it} + \zeta_{it})}_{=\epsilon_t^\Gamma} \quad (69)$$

which shows a relationship between the estimated MPL β_d and the estimated coefficient on the granular residual β_Γ . Equation (69) makes clear that as the market share of granular banks $\sum_{i=1}^K \omega_{it} \leq 1$ increase, the fit of the model also increases.

The estimated coefficient β_Γ from the granularity regression (63) in the data equals 0.517 in Table 7 which is significantly lower than the data estimate in Table 6 of dominant bank MPL of 0.959 in (61). The estimated coefficient from equation (69) in the last column of Table 7 equals 0.968 with a standard error equal to 0.170, insignificantly different from

0.959.⁴⁶

To understand this downward bias we compute the group level innovation of the change in total deposits for the Top 10 banks and for banks not in the Top 10.⁴⁷ With these group-level innovations at hand, we compute the weighted residual as in Gabaix (2011) as follows: $\Gamma_t^d = (L_{dt-1}/L_{t-1})\nu_{dt}$ and $\Gamma_t^f = (L_{ft-1}/L_{t-1})\nu_{ft}$. We then compute the correlation for years 1984-2019 (outside the financial crisis) to find that they are highly negatively correlated (the correlation is -0.925). The economics behind this difference rests on a “deposit stealing argument”. Dominant banks steal deposits from fringe banks whether through mergers or other types of competition. That would suggest that the weighted sum of dominant ν_{it} shocks for $k(i) = d$ in Γ_t^d is negatively correlated with the weighted sum of fringe ν_{it} shocks for $k(i) = f$ in ϵ_t^Γ having controlled for time, etc. Hence, our estimates of β_d in (69) using a weighted-average bank level loan growth as the dependent variable help establish the close connection between a Gabaix type regression and that of MPL regressions in (61).

⁴⁶We note that given weights ω_{it} for $k(i) = d$ in Γ_t^d covary positively with those weights in ϵ_t^Γ in (69), $\beta_d = 0.968$ is an upward biased estimate of $\beta_d = 0.959$ in (61). This occurs because the model incorrectly attributes the effect of the omitted variable to the included independent variable, causing the estimated coefficient to be larger than its true value.

⁴⁷More specifically, we let $D_{d,t} = \sum_{i=1}^K D_{it}$ (Top 10 total deposits) and $D_{f,t} = \sum_{i=K+1}^N D_{it}$ (No Top 10 total deposits). We estimate $\Delta D_{k,t} = \log(D_{k,t}) - \log(D_{k,t-1}) = \xi_k \Delta X_{kt} + \nu_{kt}$ for $k \in \{d, f\}$ to obtain ν_d and ν_f . These group level innovations allow us to capture changes in the group level deposits which are missing in the panel regressions as banks that are acquired disappear from the sample after a merger.